UFMS — Complete System Specification

Field	Value
Product	UFMS — University Financial Management System
Document type	Complete System Specification (SRS + Architecture + Features + LLD)
Date	November 2026
Classification	Proprietary

Document contents

1. System Requirements Specification (SRS)
2. Architecture Design
3. Features & Requirements
4. Low-Level Design (LLD) Workflows

Part 1 — System Requirements Specification (SRS)

University Financial Management System (UFMS)

Field	Value
Document ID	SRS-UFMS-001
Product	UFMS (University Financial Management System)
Repository	`WMSU/wmsu-platform` (`apps/finance-web` , `apps/base-api` module `finance`)
Version	1.0
Date	November 2026
Classification	Proprietary
TOR reference	VI.E.3 (finance consolidation under CIS)

---

1. Introduction

1.1 Purpose

This Software Requirements Specification defines functional and non-functional requirements for **UFMS**, the comprehensive campus financial operations system managing budgeting, government-style accounting (JEV/GL), tuition fee schedules and payment plans, cashiering and receivables, accounts payable and disbursement, procurement, point-of-sale operations — with internal APIs for USiS assessment and clearance integration.

1.2 Scope

In scope

- Budget management: plans, obligations, availability checks, monitoring, approvals, utilization tracking
- Accounting: journal entry vouchers (JEV), general ledger (GL), trial balance, bank reconciliation, fiscal period close
- Tuition and fees: fee schedules, payment policies, installment plans, assessment generation
- Cashiering and receivables: student payments, official receipt (OR) series, daily collection reports, deposit slips
- Accounts payable: supplier invoices, disbursement vouchers, check/ADA processing
- Procurement: vendor master, purchase requisition (PR), request for quotation (RFQ), purchase order (PO), inspection and acceptance report (IAR)

- Point-of-sale (POS): terminals, catalog, sales transactions, end-of-day closing; read-only stock snapshots from UIMS
- Configuration: chart of accounts (COA), fund structure, bank accounts, fiscal periods, approval workflows
- Reporting and analytics: budget vs actual, trial balance, aging reports, statement of account, dashboard KPIs
- University Platform Access SSO integration (platform code `finance`)
- Internal APIs for USiS assessment/payment handoff and clearance status

Out of scope

- Student registration, grades, admissions (owned by **USiS**)
- Transcript of records and certification issuance (owned by **TORMS**)
- Inventory master data and asset lifecycle authorship (owned by **UIMS** — UFMS POS may read stock snapshots)
- Library operations (owned by **ULMIS**)
- Document management repository (owned by **UDMS**)
- Inventing university fee policy amounts in documentation beyond configured schedules

1.3 Definitions

Term	Definition
UFMS	University Financial Management System
JEV	Journal Entry Voucher — government accounting document
GL	General Ledger — comprehensive accounting record
COA	Chart of Accounts — hierarchical account structure
OR	Official Receipt — government-issued payment receipt
DV	Disbursement Voucher — payment authorization document
IAR	Inspection and Acceptance Report — procurement delivery verification
Hard budget stop	System setting that blocks obligations when funds unavailable
Assessment	Tuition and fee calculation for a student enrollment period

1.4 References

- `WMSU/wmsu-platform/README.md`
- `apps/web-portal/public/docs/UFMS-Complete-Guide.html`
- `apps/web-portal/public/docs/UFMS-API-Specs-Manual.html`
- USiS Internal API Specification (assessment handoff)
- UIMS Stock Snapshot API Specification (POS integration)
- University Platform Access Architecture

2. Overall Description

2.1 Product Perspective

UFMS operates within the WMSU Campus Information System modular architecture:

Component	Technology	Role
Finance web	React 19 + Vite	Multi-role financial operations UI
Base API finance module	AdonisJS 7 + Lucid ORM	REST services at `/api/v1/finance/*`
Database	PostgreSQL schema `finance`	Financial system of record
Identity provider	University Platform Access	Authentication, authorization, user management
Queue system	BullMQ + Redis	Background jobs (reports, notifications, reconciliation)

Surface	Dev port	Production example
Finance web	`http://127.0.0.1:5182`	`https://finance.wmsu.edu.ph`
Base API	`http://127.0.0.1:3500`	`https://api.wmsu.edu.ph`
Identity API	`http://127.0.0.1:3411`	`https://auth.wmsu.edu.ph`

Health endpoint: `GET /health` → `{ status: 'ok', service: 'base-api', module: 'finance' }`

2.2 User Classes

Role	Primary capabilities	Access level
Cashier	Record student payments, issue official receipts, daily collection, POS operations	Payment processing
Accounting	Journals, GL, reconciliation, trial balance, period close	Full accounting
Budget Officer	Budget plans, obligations, monitoring, approval workflows	Budget oversight
Procurement Officer	PR → RFQ → PO → IAR workflows, vendor management	Procurement processes
Treasury	Disbursement vouchers, check/ADA processing, bank reconciliation	Payment disbursement
Registrar (limited)	Read-only access to student account status, clearance verification	Cross-system queries

Identity Administrator	Full UFMS access when `finance` platform granted	All UFMS permissions

Frontend permission checks provide user experience; API enforces authorization on every route.

2.3 Constraints

- Platform tile visibility requires Identity platform grant `finance`
- Internal USiS integration routes require service API key protection (browser never receives key)
- Budget availability may enforce hard-stop when configured — blocks obligations if funds insufficient
- Fee schedules are configuration data — this SRS does not invent tariff amounts
- COA structure and fiscal periods must be configured before live posting
- OR series must be activated and assigned to cashiers before receipt issuance
- Government accounting conventions (JEV, GL) follow Philippine GASB/IPSAS-aligned standards
- POS reads stock data from UIMS as snapshots — UFMS does not author inventory masters

2.4 Assumptions

- COA, fund structure, and fiscal periods seeded or configured during deployment
- USiS enrollment system calls UFMS internal APIs for assessment upsert and payment status queries
- SMTP/notification infrastructure configured for approval alerts and reminders
- University seal and authorized signatory images available for document generation
- Bank account details configured for reconciliation
- Cashiers assigned to specific OR series for accountability
- Demo/staging environments provision sample COA, students, transactions
- UIMS inventory system running when POS stock integration enabled

---

3. Functional Requirements

3.1 Authentication & Authorization

ID	Requirement	Priority
FR-AUTH-001	System shall authenticate all users via University Platform Access SSO	Must
FR-AUTH-002	System shall resolve user home path based on role (cashier → `/cashier`, accounting → `/accounting`, etc.)	Must
FR-AUTH-003	System shall enforce `finance.*` permission checks on every authenticated API endpoint	Must
FR-AUTH-004	Internal service routes shall require `X-Service-Api-Key` header validation	Must

FR-AUTH-005	Browser clients shall never receive service API keys	Must

3.2 Configuration Management

ID	Requirement	Priority
FR-CFG-001	Administrators shall configure chart of accounts (COA) with hierarchical structure	Must
FR-CFG-002	System shall support fund structure configuration (general, special, trust, etc.)	Must
FR-CFG-003	Administrators shall configure bank accounts with reconciliation parameters	Must
FR-CFG-004	Administrators shall define fiscal periods with open/closed status	Must
FR-CFG-005	System shall configure OR series with prefix, start number, end number, and cashier assignment	Must
FR-CFG-006	Administrators shall configure approval workflows for budgets, disbursements, procurement	Should
FR-CFG-007	System shall maintain audit log of configuration changes	Must

3.3 Budget Management

ID	Requirement	Priority
FR-BDG-001	Budget officers shall create budget plans with allocation by fund, account, and period	Must
FR-BDG-002	System shall support budget approval workflow (draft → submitted → approved)	Must
FR-BDG-003	System shall track obligations against approved budget	Must
FR-BDG-004	System shall check budget availability before allowing obligation posting	Must
FR-BDG-005	System shall enforce hard budget stop when configured (block if insufficient funds)	Must
FR-BDG-006	Budget officers shall monitor budget utilization with real-time reports	Must
FR-BDG-007	System shall generate budget vs actual reports by period, fund, account	Must
FR-BDG-008	System shall support budget amendments and realignments with approval	Should

3.4 Accounting

ID	Requirement	Priority

FR-ACC-001	Accountants shall create journal entry vouchers (JEV) with debit/credit lines	Must
FR-ACC-002	System shall validate JEV balance (total debit = total credit) before posting	Must
FR-ACC-003	System shall post JEV entries to general ledger (GL) with audit trail	Must
FR-ACC-004	System shall maintain GL balances by account, fund, and fiscal period	Must
FR-ACC-005	System shall generate trial balance report for specified period	Must
FR-ACC-006	Accountants shall perform bank reconciliation with imported bank statements	Must
FR-ACC-007	System shall support fiscal period close with validation and lock mechanism	Must
FR-ACC-008	System shall prevent posting to closed periods without administrator override	Must
FR-ACC-009	System shall generate financial statements (income statement, balance sheet)	Should

3.5 Tuition and Fee Management

ID	Requirement	Priority
FR-TUI-001	Administrators shall configure fee schedules by program, year level, and term	Must
FR-TUI-002	System shall define payment policies (full, installment) with due dates	Must
FR-TUI-003	System shall calculate tuition assessments based on enrolled units and fee schedules	Must
FR-TUI-004	System shall generate assessment documents linked to USiS enrollments	Must
FR-TUI-005	System shall track installment payment schedules with penalty calculations	Should
FR-TUI-006	System shall support scholarships and discounts with GL posting rules	Should

3.6 Cashiering and Receivables

ID	Requirement	Priority
FR-CSH-001	Cashiers shall look up student accounts by student number or name	Must
FR-CSH-002	System shall display outstanding assessments and payment history	Must
FR-CSH-003	Cashiers shall record student payments with payment mode (cash, check, online)	Must
FR-CSH-004	System shall reject invalid payments (unknown student, zero amount, negative amount)	Must
FR-CSH-005	System shall issue official receipt (OR) from assigned series upon payment	Must
FR-CSH-006	System shall generate OR PDF with payment details, OR number, and QR code	Must
FR-CSH-007	System shall post payment to GL automatically (debit cash, credit receivable)	Must
FR-CSH-008	Cashiers shall generate daily collection report at end of shift	Must

FR-CSH-009	System shall create deposit slip for bank remittance	Must
FR-CSH-010	System shall track OR series usage and alert when nearing end	Should

3.7 Accounts Payable and Disbursement

ID	Requirement	Priority
FR-APY-001	Accountants shall register supplier invoices with amount, due date, and terms	Must
FR-APY-002	System shall create disbursement vouchers (DV) linked to invoices or obligations	Must
FR-APY-003	System shall support DV approval workflow with authorization limits	Must
FR-APY-004	Treasury shall process approved DVs with check or ADA (advice to debit account)	Must
FR-APY-005	System shall post disbursement to GL (debit expense/asset, credit payable/cash)	Must
FR-APY-006	System shall track check status (issued, cleared, stale, cancelled)	Should
FR-APY-007	System shall generate aging report for accounts payable	Must

3.8 Procurement

ID	Requirement	Priority
FR-PRC-001	System shall maintain supplier/vendor master with contact and tax details	Must
FR-PRC-002	Requestors shall create purchase requisitions (PR) with item details and justification	Must
FR-PRC-003	System shall support PR approval workflow	Must
FR-PRC-004	Procurement officers shall issue request for quotation (RFQ) to suppliers	Should
FR-PRC-005	Procurement officers shall award supplier and create purchase order (PO)	Must
FR-PRC-006	System shall track PO status (issued, partially received, fully received, cancelled)	Must
FR-PRC-007	Receiving officers shall create inspection and acceptance report (IAR) upon delivery	Must
FR-PRC-008	System shall link IAR to PO and trigger accounts payable invoice creation	Must
FR-PRC-009	System shall enforce budget availability check before PO creation	Must

3.9 Point-of-Sale (POS)

ID	Requirement	Priority
FR-POS-001	Cashiers shall configure POS terminals with assigned location and cashier	Must
FR-POS-002	System shall maintain POS catalog with items and prices	Must
FR-POS-003	System shall query UIMS for real-time stock availability (read-only snapshot)	Should

FR-POS-004	Cashiers shall build sale transaction with multiple line items	Must
FR-POS-005	System shall calculate sale total with taxes/discounts if configured	Should
FR-POS-006	System shall complete sale with payment and issue OR/receipt	Must
FR-POS-007	System shall post POS sale to GL (debit cash, credit revenue)	Must
FR-POS-008	Cashiers shall perform end-of-day (EOD) closing with summary report	Must
FR-POS-009	System shall prevent further sales on closed terminal until new day started	Must

3.10 Internal API for USiS Integration

ID	Requirement	Priority
FR-INT-001	USiS shall upsert student assessments via internal API (service key protected)	Must
FR-INT-002	System shall accept assessment data (student number, term, fee line items) and generate assessment document	Must
FR-INT-003	USiS shall query payment status via internal API	Must
FR-INT-004	System shall return payment status (pending, partial, verified) with amounts	Must
FR-INT-005	USiS/TORMS shall query financial clearance status via internal API	Must
FR-INT-006	System shall return clearance status (cleared if no outstanding balance, otherwise balance amount)	Must

3.11 Dashboard and Reporting

ID	Requirement	Priority
FR-DSH-001	Finance dashboard shall display KPIs: total collections, outstanding receivables, disbursements, budget utilization	Must
FR-DSH-002	Dashboard shall show recent transactions and pending approvals	Should
FR-RPT-001	System shall generate trial balance report by period and fund	Must
FR-RPT-002	System shall generate budget vs actual report with variance analysis	Must
FR-RPT-003	System shall generate accounts receivable aging report	Must

FR-RPT-004	System shall generate accounts payable aging report	Must
FR-RPT-005	System shall generate statement of account for students	Must
FR-RPT-006	System shall generate daily collection report by cashier	Must
FR-RPT-007	System shall export all reports in CSV, Excel, and PDF formats	Must
FR-RPT-008	System shall log all report exports with user and timestamp for audit	Must

---

4. External Interface Requirements

4.1 Integration Interfaces

Interface	Direction	Protocol	Purpose
University Platform Access	Inbound	OIDC/JWKS	SSO authentication, JWT verification, role resolution
USiS Enrollment API	Inbound	REST+JSON	Assessment upsert, payment status query, clearance verification
TORMS Records API	Inbound	REST+JSON	Clearance verification for TOR requests
UIMS Inventory API	Outbound (optional)	REST+JSON	POS stock availability snapshots
SMTP/SMS Gateway	Outbound	SMTP/HTTP	Approval notifications, payment reminders
Bank API	Outbound (optional)	SFTP/API	Bank statement import for reconciliation
Object Storage	Outbound	S3-compatible	OR PDF archival, report storage

4.2 API Specifications

Finance API base: `/api/v1/finance/*`

Internal service routes: `/api/v1/finance/internal/*` (service key protected)

OpenAPI documentation: Available at `/api/v1/finance/docs` (Swagger UI)

Authentication: Bearer JWT from University Platform Access in `Authorization` header

4.3 User Interface Requirements

- Responsive design supporting desktop (≥1280px), tablet (768-1023px), and mobile (360-767px)
- Browser compatibility: Chrome/Edge (latest 2), Firefox (latest 2), Safari (latest 2)
- Accessibility: WCAG 2.1 Level AA for public-facing payment status pages
- Print-optimized views for OR, financial statements, reports
- Real-time budget availability indicators
- Bulk transaction processing interfaces for accounting
- POS terminal optimized for touch/barcode scanner input

---

5. Non-Functional Requirements

5.1 Performance

ID	Requirement	Target
NFR-PERF-001	API response time (p95) for standard queries	≤ 2 seconds
NFR-PERF-002	Payment recording and OR generation	≤ 3 seconds
NFR-PERF-003	Budget availability check	≤ 1 second
NFR-PERF-004	Health check endpoint response time	≤ 500 ms
NFR-PERF-005	Trial balance generation for 10,000 transactions	≤ 30 seconds

5.2 Scalability

ID	Requirement
NFR-SCAL-001	Support ≥20,000 student accounts
NFR-SCAL-002	Support ≥100,000 transactions per academic year
NFR-SCAL-003	Horizontal scaling of API replicas
NFR-SCAL-004	Background job queue processing with worker pools

5.3 Security

ID	Requirement
NFR-SEC-001	TLS 1.2+ required for all client-server communication
NFR-SEC-002	RBAC enforcement on every authenticated API endpoint using `finance.*` permissions
NFR-SEC-003	Service-to-service communication via API keys for internal routes
NFR-SEC-004	Financial data encryption at rest for sensitive fields (bank accounts, OR numbers)

NFR-SEC-005	Audit logging of all financial transactions, posting, and configuration changes
NFR-SEC-006	CORS allowlist limited to authorized web client origins
NFR-SEC-007	Non-root container execution in production
NFR-SEC-008	Budget and disbursement approval workflows with authorization limits
NFR-SEC-009	OR series accountability tracking by cashier
NFR-SEC-010	Password/MFA enforcement for privileged financial roles

5.4 Reliability

ID	Requirement
NFR-REL-001	System availability ≥99.5% during business hours
NFR-REL-002	Automated database backups (daily incremental, weekly full)
NFR-REL-003	Point-in-time recovery capability ≤15 minutes RPO
NFR-REL-004	Graceful degradation when USiS/UIMS temporarily unavailable
NFR-REL-005	Posted GL entries immutable — corrections via reversing JEV only
NFR-REL-006	Fiscal period close prevents accidental posting to closed periods

5.5 Maintainability

ID	Requirement
NFR-MAIN-001	Modular domain architecture with separation of concerns
NFR-MAIN-002	Comprehensive API documentation (OpenAPI 3.0+)
NFR-MAIN-003	Structured logging with correlation IDs
NFR-MAIN-004	Database schema migrations versioned and reversible
NFR-MAIN-005	COA and configuration hot-reloading without deployment

5.6 Compliance

ID	Requirement
NFR-COMP-001	RA 10173 (Data Privacy Act) compliance for student financial data
NFR-COMP-002	Philippine government accounting standards alignment (GASB/IPSAS)
NFR-COMP-003	Audit trail retention ≥7 years per government records management
NFR-COMP-004	OR series accountability and reconciliation

NFR-COMP-005	Financial statement accuracy and timeliness

6. Verification Criteria

6.1 Functional Verification

Scenario	Expected result	Test type
Cashier login via SSO	Redirect to `/cashier` dashboard	Integration
Record student payment	Payment saved, OR issued, GL posted	E2E
Record payment with invalid student number	HTTP 422 rejection	Unit
Record payment with zero amount	HTTP 422 rejection	Unit
Budget obligation exceeds available with hard stop	HTTP 422 rejection	Unit
Budget obligation without hard stop	Allowed with warning	Unit
Post JEV with unbalanced debit/credit	HTTP 422 validation error	Unit
USiS upsert assessment via internal API	Assessment created, linked to student	Integration
USiS query payment status	Returns pending/partial/verified accurately	Integration
Query clearance with outstanding balance	Returns not cleared with balance amount	Integration
Query clearance with zero balance	Returns cleared	Integration
POS query UIMS stock	Returns available quantity	Integration (optional)
Generate trial balance	Balanced report with all accounts	Integration
Fiscal period close	Period locked, posting prevented	E2E

6.2 Non-Functional Verification

Scenario	Expected result	Test type
100 concurrent payment recordings	p95 response ≤3s, no errors	Load test
Trial balance for 10K transactions	Completes in ≤30s	Performance test
Unauthorized access to internal API	HTTP 401/403	Security test
Database failover	Recovery within 15 minutes	DR drill

7. Document Control

Version	Date	Description
1.0	November 2026	Complete System Specification from wmsu-platform finance module

Related Design Documents

- **UFMS-Architecture.md** — Architecture design (containers, domains, security, deployment)
- **UFMS-Features-Requirements.md** — Feature catalog + requirements traceability
- **UFMS-LLD.md** — Low-level design workflows (cashiering, budget, procurement, disbursement, POS, USiS integration)

End of SRS — UFMS

Part 2 — Architecture Design

UFMS — University Financial Management System Architecture

Field	Value
Document ID	AD-UFMS-001
Product	UFMS
Companions	UFMS-SRS.md · UFMS-Features-Requirements.md · UFMS-LLD.md
Date	November 2026

---

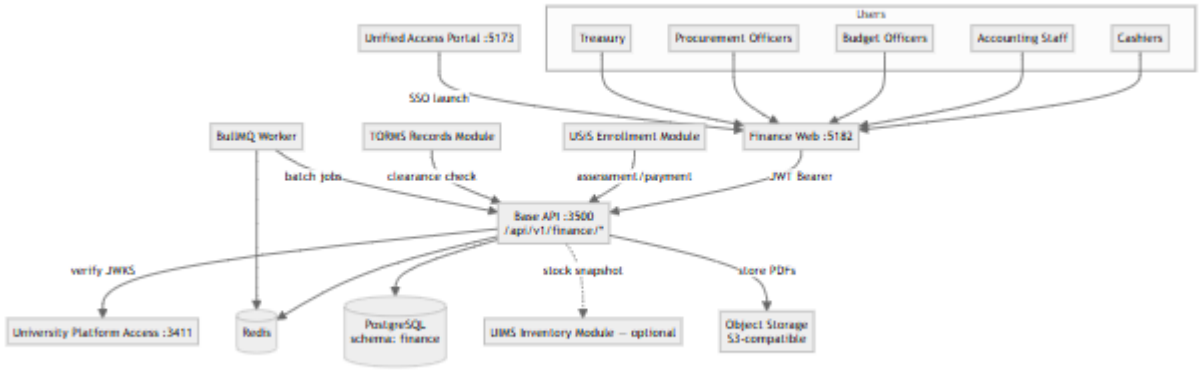
1. Purpose

Architecture design for UFMS within WMSU CIS modular monolith — covering system context, container architecture, domain module organization, security architecture, deployment topology, integration patterns, and quality attributes for university financial management.

---

2. System Context

UFMS integrates with Identity for SSO, USiS for assessment/payment handoff, and optional UIMS for POS stock:



Key relationships:

- Financial staff authenticate via Portal SSO
- All UI interactions flow through Base API finance module

- USiS enrollment system calls UFMS internal APIs for assessment upsert and payment verification
- TORMS records system calls UFMS for clearance checks
- UFMS may query UIMS for POS stock availability (optional integration)
- Background jobs (reports, reconciliation, notifications) run via BullMQ worker
- OR PDFs and financial reports archived to object storage

---

3. Container View

Container	Technology	Port	Purpose
Finance web	React 19 + Vite	5182	Multi-role financial operations UI
Base API (finance module)	AdonisJS 7 + Lucid ORM	3500	Domain services, controllers, business logic
PostgreSQL	PostgreSQL 15+	5432	Schema `finance` — financial system of record
Redis	Redis 7+	6379	Cache, queue backend, rate limiting
BullMQ Worker	Node.js + BullMQ	N/A	Background jobs (reports, reconciliation)
Object Storage	S3-compatible	N/A	OR PDF archival, report storage
University Platform Access	Identity service	3411	Authentication, authorization
USiS Enrollment	CIS module	3500	Assessment/payment client

Deployment notes:

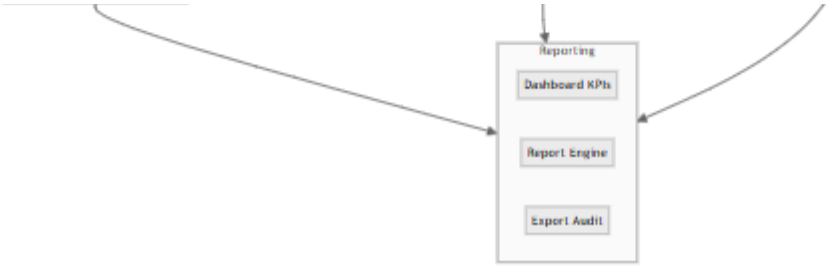
- Finance web served statically from CDN or Vite dev server
- Base API runs in containerized Node.js (non-root)
- PostgreSQL schema `finance` isolated within `wmsu_cis` database
- Redis persistent (AOF + RDB) for queue durability
- Worker pool for concurrent job processing

---

4. Domain Module Architecture

UFMS finance module organized by financial operation domains:


POS
POS Terminals
POS Catalog
Sales Transactions
End-of-Day Closing
optional
Integration
USIS Assessment/Payment
TORMS Clearance
UIMS Stock Snapshot
Core
Foundation
Org, Settings
Chart of Accounts
Fund Structure
Fiscal Periods
RBAC
Roles & Permissions
Procurement
Vendor Master
Purchase Requisitions
Request for Quotation
Purchase Orders
Inspection & Acceptance
Tuition
Fee Schedules
Payment Policies
Student Assessments
Scholarships & Discounts
Budget
Budget Plans
Obligations
Availability Checks
Monitoring & Reports
APAndDisbursement
Supplier Invoices
Disbursement Vouchers
Checks / ADA Processing
AP Aging
Cashiering
Student Payments
OR Series Management
Daily Collection
Deposit Slips
Accounting
Journal Entry Vouchers
General Ledger
Trial Balance
Bank Reconciliation
Period Close



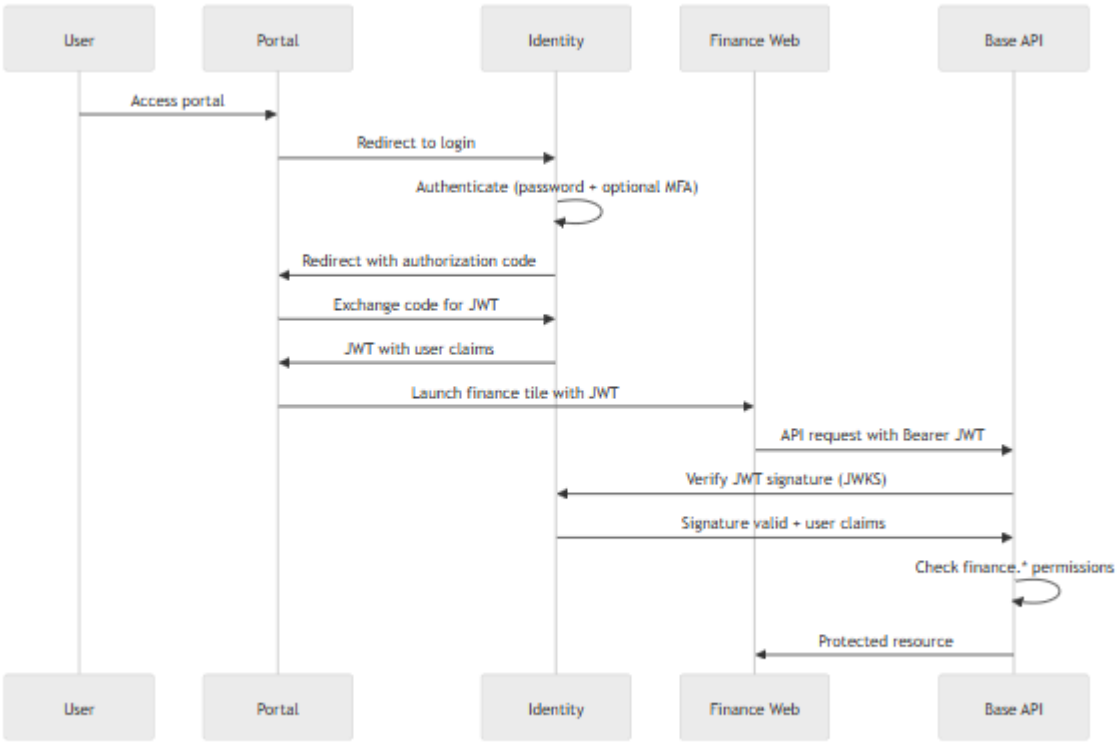
Domain responsibilities:

Domain	Key entities	Services
Core	Organization, Campus, COA, Fund, FiscalPeriod, Settings	Foundation config, COA management
Budget	BudgetPlan, BudgetLine, Obligation	Budget planning, obligation tracking, availability checks
Accounting	JournalEntry, GLAccount, GLBalance, BankReconciliation	JEV posting, GL maintenance, trial balance, period close
Tuition	FeeSchedule, PaymentPolicy, Assessment, Scholarship	Fee configuration, assessment generation
Cashiering	Payment, OfficialReceipt, ORSeries, CollectionReport	Payment processing, OR issuance, daily collection
AP & Disbursement	Invoice, DisbursementVoucher, Check, ADA	Supplier invoices, payment disbursement
Procurement	Vendor, PurchaseRequisition, PurchaseOrder, InspectionReport	Procurement lifecycle
POS	POSTerminal, POSCatalogItem, Sale, SaleLine	Point-of-sale operations
Integration	AssessmentSync, PaymentStatusQuery, ClearanceCheck, StockSnapshot	External system integration
Reporting	Report, ReportExport, DashboardMetric	Analytics, export audit

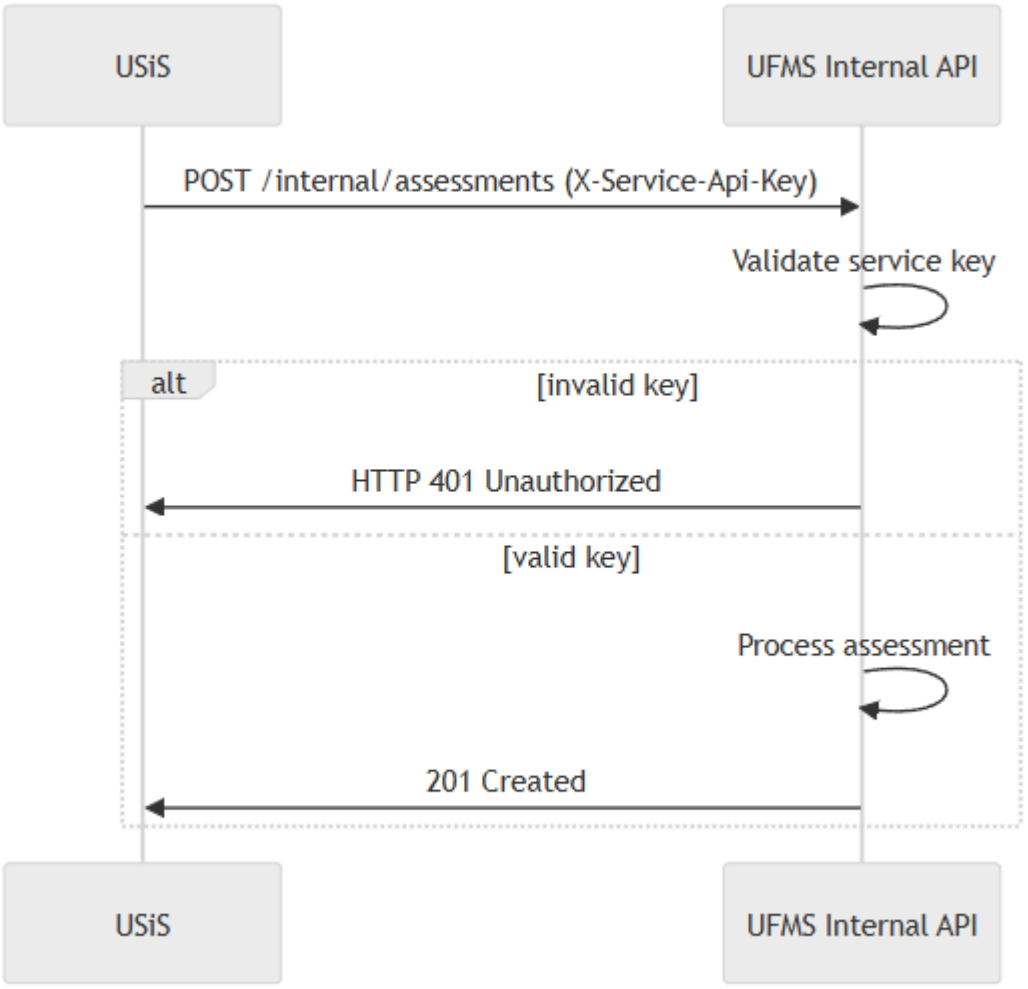
---

5. Security Architecture

5.1 Authentication Flow



Internal API flow (USiS):



Security controls:

Layer	Control	Implementation
Transport	TLS 1.2+	Reverse proxy termination
Authentication	OIDC/JWT	Identity service RS256 tokens
Authorization	RBAC	`finance.*` permission checks
Internal APIs	Service keys	`X-Service-Api-Key` header
Financial data	Encryption at rest	Sensitive fields encrypted (bank accounts, OR)
Approval workflows	Authorization limits	Role-based approval authority
OR accountability	Cashier assignment	OR series tracked by cashier

5.2 Permission Model

UFMS permission categories:

- `finance.cashier.*` — Payment processing, OR issuance
- `finance.accounting.*` — JEV, GL, reconciliation, period close
- `finance.budget.*` — Budget planning, obligations, monitoring
- `finance.procurement.*` — PR, PO, IAR workflows
- `finance.treasury.*` — Disbursement vouchers, checks/ADA
- `finance.pos.*` — POS operations
- `finance.reports.*` — Report generation and export
- `finance.config.*` — System configuration

Role mappings:

- `cashier` → `finance.cashier.*`, `finance.pos.*`
- `accountant` → `finance.accounting.*`, `finance.reports.view`
- `budget_officer` → `finance.budget.*`, `finance.reports.budget`
- `procurement_officer` → `finance.procurement.*`
- `treasurer` → `finance.treasury.*`, `finance.accounting.view`
- `finance_admin` → All `finance.*` permissions

---

6. Data Architecture

6.1 Schema Organization

PostgreSQL schema `finance` contains:

Core tables:

- `organizations`, `campuses`, `settings`
- `chart_of_accounts`, `funds`, `fiscal_periods`

Budget:

- `budget_plans`, `budget_lines`, `obligations`

Accounting:

- `journal_entries`, `journal_entry_lines`, `gl_accounts`, `gl_balances`
- `bank_accounts`, `bank_reconciliations`

Tuition:

- `fee_schedules`, `payment_policies`, `assessments`, `assessment_lines`
- `scholarships`, `scholarship_grants`

Cashiering:

- `payments`, `official_receipts`, `or_series`
- `collection_reports`, `deposit_slips`

AP & Disbursement:

- `vendors`, `invoices`, `disbursement_vouchers`
- `checks`, `ada_records`

Procurement:

- `purchase_requisitions`, `purchase_orders`, `inspection_reports`

POS:

- `pos_terminals`, `pos_catalog_items`, `sales`, `sale_lines`

Integration:

- `assessment_syncs` (USiS)
- `payment_status_queries` (USiS)
- `clearance_checks` (TORMS)
- `stock_snapshots` (UIMS)

Reporting:

- `report_exports` (audit log)

6.2 Integration Strategy

USiS → UFMS integration:

- **Assessment upsert:** USiS posts assessment data to UFMS when enrollment submitted
- **Payment status query:** USiS queries UFMS for real-time payment verification
- **Clearance query:** USiS/TORMS check student financial clearance status
- **Authentication:** Service API key in `X-Service-Api-Key` header
- **Frequency:** Real-time on-demand

UFMS → UIMS integration (optional POS stock):

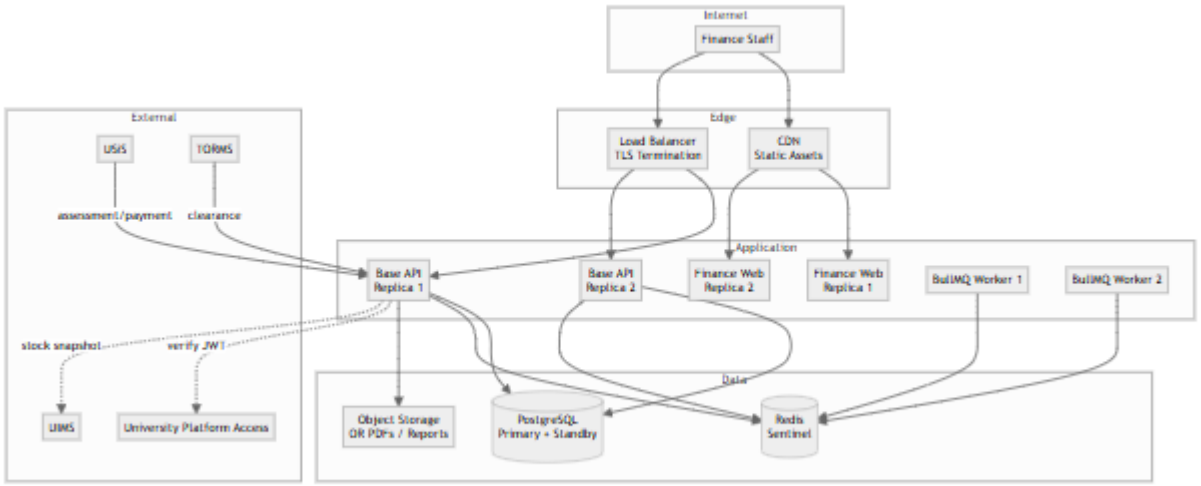
- **Stock query:** UFMS queries UIMS for item availability during POS transaction
- **Pattern:** Read-only snapshot, no write-back to UIMS

- **Frequency:** Real-time during POS operations

---

7. Deployment Topology

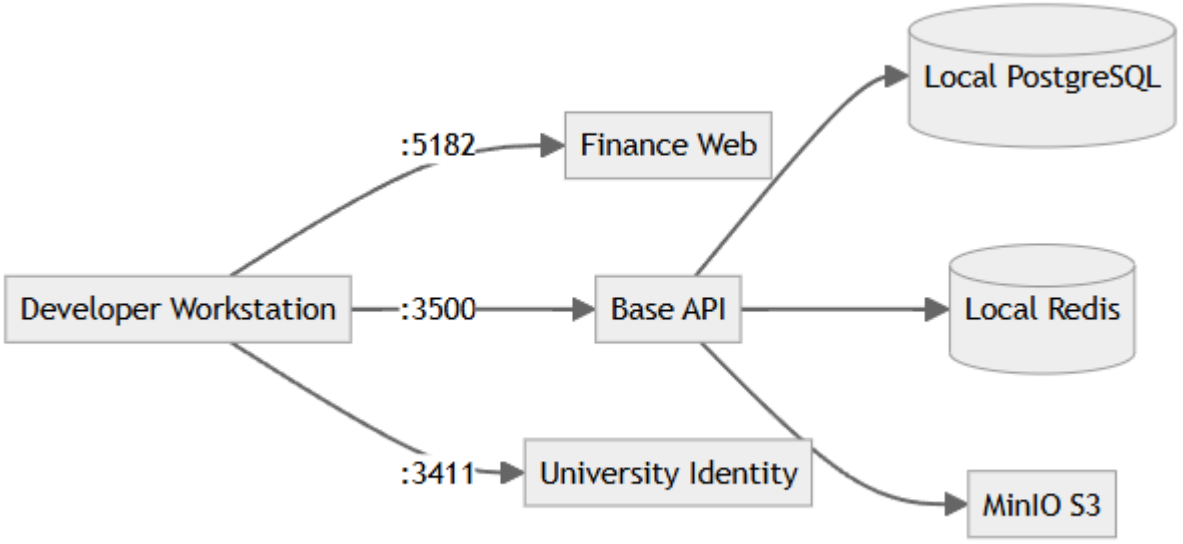
7.1 Production Architecture



Sizing guidance:

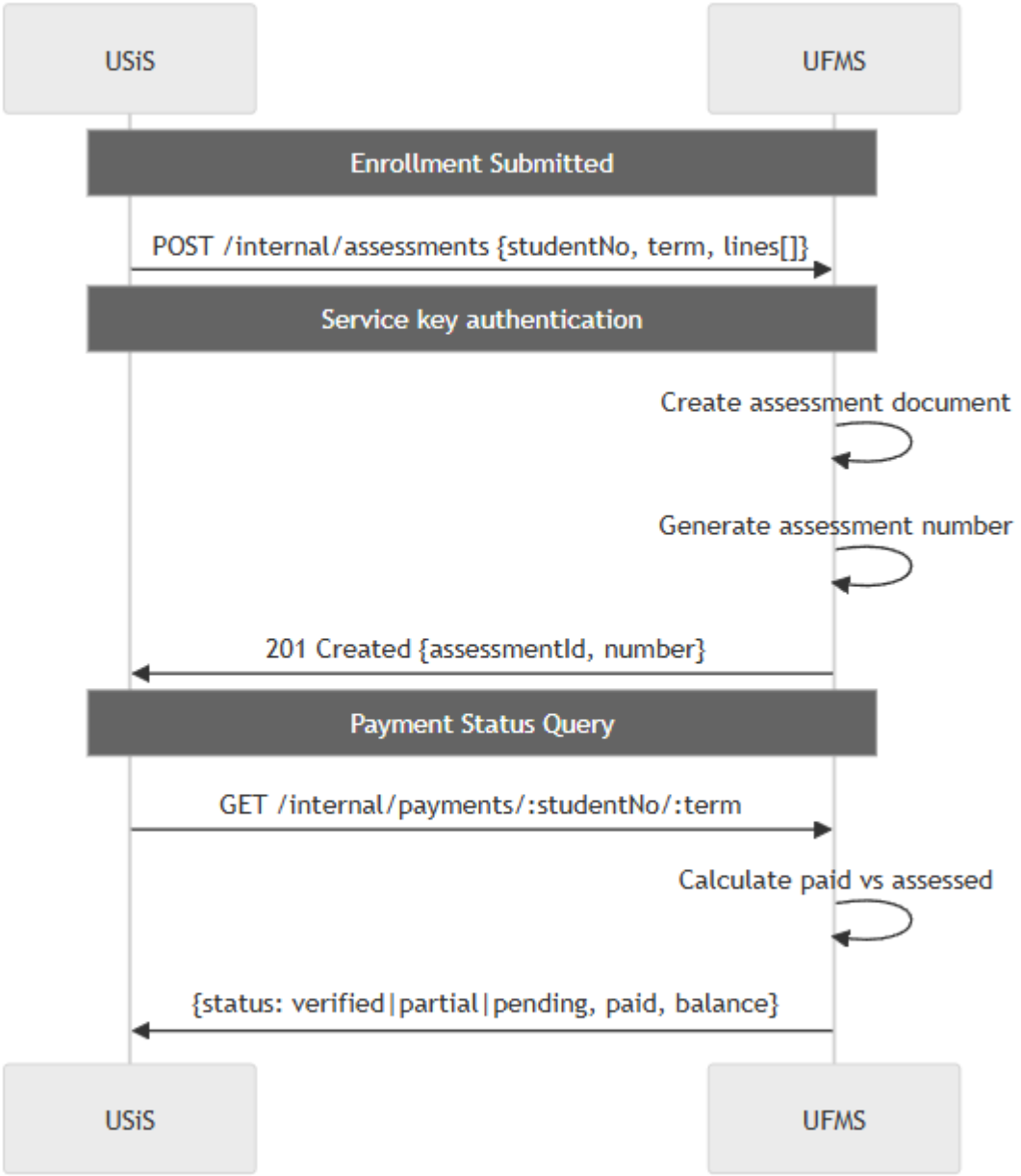
- API: 2–3 replicas, 2 vCPU / 4 GB RAM each
- Worker: 2 replicas, 1 vCPU / 2 GB RAM each
- PostgreSQL: 4–8 vCPU / 16–32 GB RAM (financial workload)
- Redis: 2 vCPU / 4 GB RAM
- Object Storage: S3-compatible managed service

7.2 Local Development

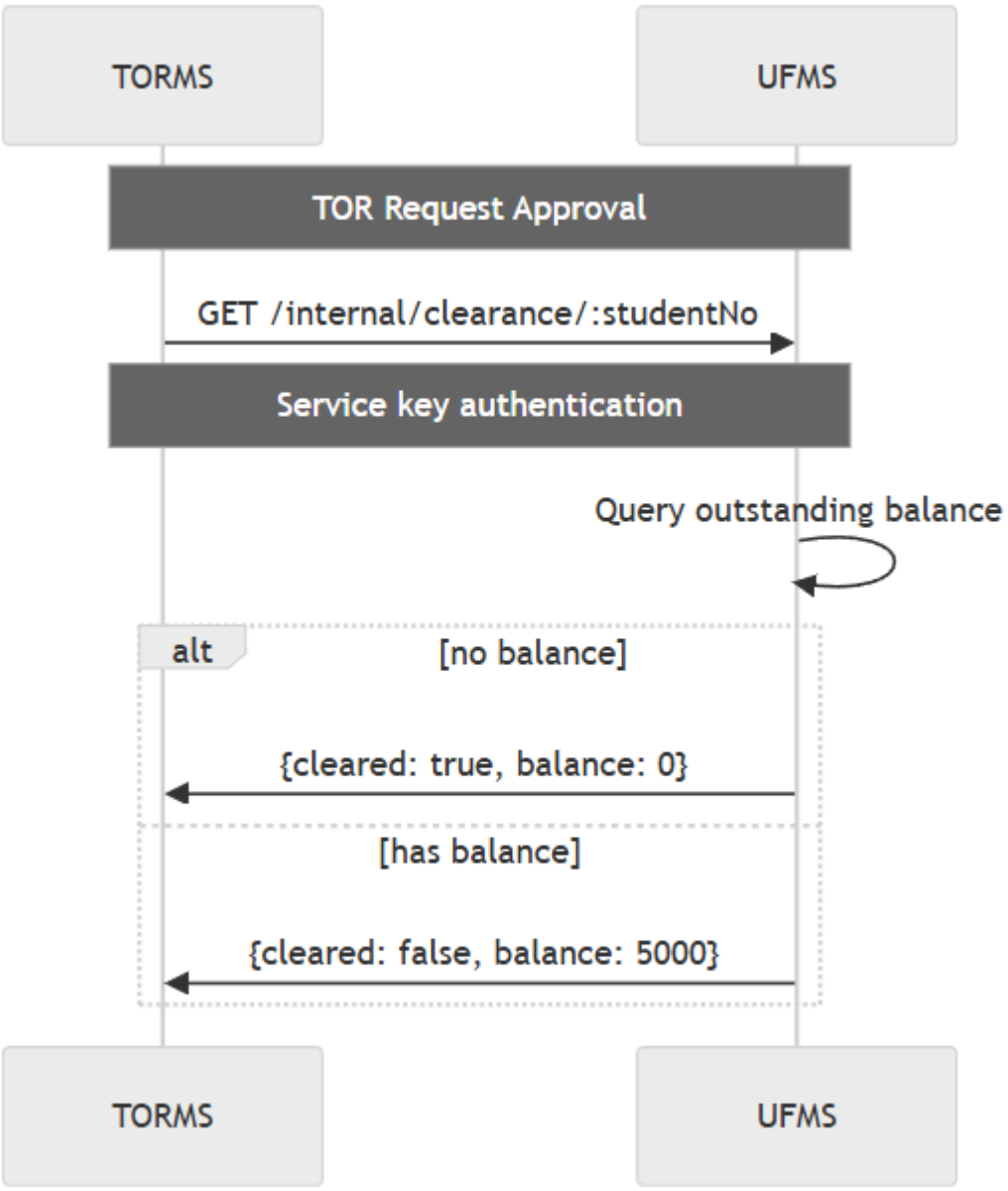


8. Integration Architecture

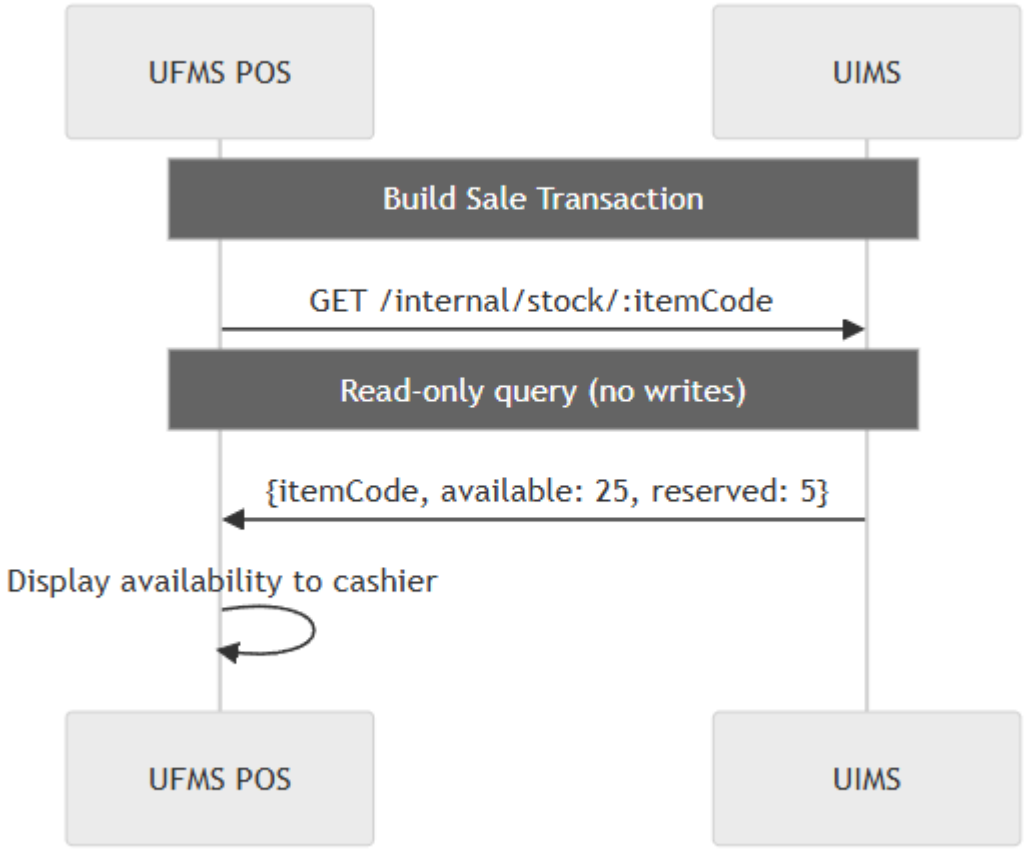
8.1 USiS Assessment Handoff



8.2 TORMS Clearance Check



8.3 UIMS Stock Snapshot (Optional POS)



---

9. Quality Attributes

9.1 Performance

Optimization strategies:

- Database indexing on student_no, account_code, fiscal_period, status
- Redis caching for COA structure, fee schedules, OR series
- Background report generation (async job queue) for large datasets
- Lazy loading in finance web UI
- Paginated API responses (default 25, max 100)
- Read replicas for reporting queries

Monitoring:

- APM (Sentry/OpenTelemetry) for request tracing
- Payment processing time metrics
- Budget availability check latency
- GL posting performance

- Report generation duration

9.2 Scalability

Horizontal scaling:

- Stateless API design
- Load balancer with health checks
- Worker pool for concurrent job processing
- Database connection pooling (PgBouncer)

Data growth:

- Partitioning for large tables (journal_entry_lines, payments) by fiscal year
- Archival strategy for old fiscal periods (closed > 7 years)

9.3 Reliability

Failure handling:

- Graceful degradation when USiS/UIMS unavailable (cache last-known status, warn users)
- Retry logic for internal API calls (exponential backoff)
- Circuit breaker for external service calls
- Immutable posted GL entries — corrections via reversing JEV only
- Fiscal period close prevents accidental posting

Data integrity:

- Foreign key constraints
- Transaction management with rollback
- JEV debit/credit balance validation
- Budget availability checks before obligations
- OR series sequence integrity

9.4 Maintainability

Code organization:

- Domain-driven design with bounded contexts
- Service layer abstraction
- Shared validation schemas
- OpenAPI documentation

Operational:

- Structured JSON logging
- Feature flags
- Blue-green deployment
- Automated schema migrations

- COA hot-reloading

9.5 Compliance

Data privacy (RA 10173):

- Student financial data access restricted by role
- Audit logging of payment access and modifications
- Retention policies configurable

Financial accountability:

- Posted GL entries immutable (reversing JEV for corrections)
- OR series accountability by cashier
- 7-year audit trail retention
- Budget vs actual transparency
- Fiscal period integrity

End of Architecture Design — UFMS

Part 3 — Features & Requirements

UFMS — University Financial Management System Features & Requirements

Field	Value
Document ID	FEAT-UFMS-001
Product	UFMS
Companions	UFMS-SRS.md · UFMS-LLD.md
Date	November 2026

---

1. Feature Summary

UFMS provides comprehensive campus financial operations management:

Area	Features	Status
Authentication	University Platform Access SSO, role-based routing	Delivered
Configuration	COA, funds, fiscal periods, OR series, approval workflows	Delivered
Budget	Planning, obligations, availability checks, monitoring, reports	Delivered
Accounting	JEV, GL, trial balance, bank reconciliation, period close	Delivered
Tuition	Fee schedules, payment policies, assessments, scholarships	Delivered
Cashiering	Student payments, OR issuance, daily collection, deposit slips	Delivered
AP & Disbursement	Invoices, disbursement vouchers, checks/ADA, aging reports	Delivered
Procurement	Vendor master, PR, RFQ, PO, IAR workflows	Delivered
POS	Terminals, catalog, sales, EOD closing, UIMS stock integration	Delivered
USiS Integration	Assessment upsert, payment status query, clearance verification	Delivered
Dashboard	KPIs, recent transactions, pending approvals	Delivered
Reporting	Trial balance, budget vs actual, aging, SOA, export audit	Delivered

---

2. Feature Catalog

Feature ID	Feature	Description	Priority	Actors

F-AUTH-01	SSO Authentication	University Platform Access OIDC/JWT	Must	All
F-AUTH-02	Role Routing	Home path by role (cashier/accounting/budget/procurement)	Must	All
F-AUTH-03	Permission Enforcement	`finance.*` checks on all endpoints	Must	System
F-AUTH-04	Service Key Protection	Internal API routes require service key	Must	System
F-CFG-01	COA Management	Configure chart of accounts with hierarchy	Must	Admin
F-CFG-02	Fund Structure	Configure funds (general, special, trust)	Must	Admin
F-CFG-03	Bank Accounts	Configure bank accounts with reconciliation parameters	Must	Admin
F-CFG-04	Fiscal Periods	Define periods with open/closed status	Must	Admin
F-CFG-05	OR Series Management	Configure OR series with cashier assignment	Must	Admin
F-CFG-06	Approval Workflows	Configure approval workflows with limits	Should	Admin
F-CFG-07	Config Audit Log	Log configuration changes	Must	System
F-BDG-01	Budget Planning	Create budget plans with allocations	Must	Budget Officer
F-BDG-02	Budget Approval	Workflow: draft → submitted → approved	Must	Budget Officer
F-BDG-03	Obligation Tracking	Track obligations against budget	Must	System
F-BDG-04	Availability Check	Verify funds before obligation	Must	System
F-BDG-05	Hard Budget Stop	Block obligations if insufficient funds (configurable)	Must	System
F-BDG-06	Budget Monitoring	Real-time utilization reports	Must	Budget Officer
F-BDG-07	Budget vs Actual	Generate variance reports	Must	Budget Officer
F-BDG-08	Budget Amendments	Support realignments with approval	Should	Budget Officer

F-ACC-01	JEV Creation	Create journal entry vouchers	Must	Accountant
F-ACC-02	JEV Validation	Validate debit/credit balance	Must	System
F-ACC-03	GL Posting	Post JEV to general ledger	Must	System
F-ACC-04	GL Balances	Maintain balances by account/fund/period	Must	System
F-ACC-05	Trial Balance	Generate trial balance report	Must	Accountant
F-ACC-06	Bank Reconciliation	Import and reconcile bank statements	Must	Accountant
F-ACC-07	Period Close	Close fiscal period with lock	Must	Accountant
F-ACC-08	Closed Period Protection	Prevent posting to closed periods	Must	System
F-ACC-09	Financial Statements	Generate income statement, balance sheet	Should	Accountant
F-TUI-01	Fee Schedules	Configure fees by program/level/term	Must	Admin
F-TUI-02	Payment Policies	Define payment policies with due dates	Must	Admin
F-TUI-03	Assessment Calculation	Calculate tuition from enrolled units	Must	System
F-TUI-04	Assessment Generation	Generate assessment documents	Must	System
F-TUI-05	Installment Tracking	Track installment schedules with penalties	Should	System
F-TUI-06	Scholarships & Discounts	Support scholarship application to assessments	Should	Admin
F-CSH-01	Student Account Lookup	Look up by student number or name	Must	Cashier
F-CSH-02	Payment History Display	Show assessments and payment history	Must	Cashier
F-CSH-03	Payment Recording	Record payments with payment mode	Must	Cashier
F-CSH-04	Payment Validation	Reject invalid payments	Must	System

F-CSH-05	OR Issuance	Issue official receipt from assigned series	Must	System
F-CSH-06	OR PDF Generation	Generate OR PDF with QR code	Must	System
F-CSH-07	GL Auto-Posting	Post payment to GL automatically	Must	System
F-CSH-08	Daily Collection Report	Generate end-of-shift report	Must	Cashier
F-CSH-09	Deposit Slip Creation	Create bank remittance deposit slip	Must	Cashier
F-CSH-10	OR Series Alert	Alert when series nearing end	Should	System
F-APY-01	Invoice Registration	Register supplier invoices	Must	Accountant
F-APY-02	DV Creation	Create disbursement vouchers	Must	Accountant
F-APY-03	DV Approval	Workflow with authorization limits	Must	Treasury
F-APY-04	Check/ADA Processing	Process approved DVs	Must	Treasury
F-APY-05	Disbursement GL Posting	Post disbursement to GL	Must	System
F-APY-06	Check Status Tracking	Track check lifecycle	Should	Treasury
F-APY-07	AP Aging Report	Generate aging report	Must	Accountant
F-PRC-01	Vendor Master	Maintain supplier/vendor records	Must	Procurement
F-PRC-02	PR Creation	Create purchase requisitions	Must	Requestor
F-PRC-03	PR Approval	Approval workflow	Must	Procurement
F-PRC-04	RFQ Issuance	Issue request for quotation	Should	Procurement
F-PRC-05	PO Creation	Award supplier and create PO	Must	Procurement
F-PRC-06	PO Status Tracking	Track PO lifecycle	Must	Procurement

F-PRC-07	IAR Creation	Create inspection and acceptance report	Must	Receiving
F-PRC-08	IAR-to-AP Link	Link IAR to PO and create AP invoice	Must	System
F-PRC-09	Budget Check for PO	Verify budget before PO	Must	System
F-POS-01	Terminal Configuration	Configure POS terminals	Must	Admin
F-POS-02	POS Catalog	Maintain POS catalog with prices	Must	Admin
F-POS-03	Stock Availability	Query UIMS for stock (optional)	Should	System
F-POS-04	Sale Transaction	Build sale with multiple items	Must	Cashier
F-POS-05	Sale Calculation	Calculate total with taxes/discounts	Should	System
F-POS-06	Sale Completion	Complete sale with OR issuance	Must	Cashier
F-POS-07	POS GL Posting	Post sale to GL	Must	System
F-POS-08	EOD Closing	End-of-day closing with summary	Must	Cashier
F-POS-09	Closed Terminal Protection	Prevent sales on closed terminal	Must	System
F-INT-01	Assessment Upsert	USiS posts assessments via internal API	Must	System
F-INT-02	Assessment Document	Generate assessment from USiS data	Must	System
F-INT-03	Payment Status Query	USiS queries payment status	Must	System
F-INT-04	Payment Status Response	Return pending/partial/verified status	Must	System
F-INT-05	Clearance Query	TORMS/USiS query financial clearance	Must	System
F-INT-06	Clearance Response	Return cleared status with balance	Must	System
F-DSH-01	Finance Dashboard	Display KPIs, recent transactions, pending approvals	Must	All Finance

F-DSH-02	Dashboard Widgets	Collections, receivables, disbursements, budget utilization	Should	All Finance
F-RPT-01	Trial Balance Report	By period and fund	Must	Accountant
F-RPT-02	Budget vs Actual Report	With variance analysis	Must	Budget Officer
F-RPT-03	AR Aging Report	Accounts receivable aging	Must	Cashier, Accountant
F-RPT-04	AP Aging Report	Accounts payable aging	Must	Accountant
F-RPT-05	Statement of Account	Student SOA	Must	Cashier
F-RPT-06	Daily Collection Report	By cashier	Must	Cashier
F-RPT-07	Report Export	CSV, Excel, PDF formats	Must	All Finance
F-RPT-08	Export Audit Log	Log exports with user/timestamp	Must	System

---

3. Requirements Traceability Matrix

3.1 Functional Requirements Mapping

Requirement ID	Requirement	Related Features	Priority	Implementation Status
FR-AUTH-001..005	Authentication and authorization	F-AUTH-01..04	Must	Delivered
FR-CFG-001..007	Configuration management	F-CFG-01..07	Must/Should	Delivered
FR-BDG-001..008	Budget management	F-BDG-01..08	Must/Should	Delivered
FR-ACC-001..009	Accounting	F-ACC-01..09	Must/Should	Delivered
FR-TUI-001..006	Tuition and fees	F-TUI-01..06	Must/Should	Delivered
FR-CSH-001..010	Cashiering and receivables	F-CSH-01..10	Must/Should	Delivered
FR-APY-001..007	Accounts payable and disbursement	F-APY-01..07	Must/Should	Delivered
FR-PRC-001..009	Procurement	F-PRC-01..09	Must/Should	Delivered

FR-POS-001..009	Point-of-sale	F-POS-01..09	Must/Should	Delivered
FR-INT-001..006	USiS integration (internal APIs)	F-INT-01..06	Must	Delivered
FR-DSH-001..002	Dashboard	F-DSH-01..02	Must/Should	Delivered
FR-RPT-001..008	Reporting	F-RPT-01..08	Must	Delivered

3.2 Non-Functional Requirements Mapping

NFR Category	Requirement ID	Related Features	Implementation Approach
Performance	NFR-PERF-001..005	All features	Database indexing, Redis caching, async jobs, pagination
Scalability	NFR-SCAL-001..004	All features	Horizontal API scaling, worker pools, table partitioning
Security	NFR-SEC-001..010	F-AUTH-*, F-CFG-*, F-CSH-05	TLS, RBAC, service keys, encryption, audit logging, OR accountability
Reliability	NFR-REL-001..006	All features	HA deployment, backups, DR procedures, immutable GL, period close
Maintainability	NFR-MAIN-001..005	All features	Modular architecture, OpenAPI, structured logs, versioned migrations
Compliance	NFR-COMP-001..005	Financial data	RA 10173, GASB/IPSAS alignment, 7-year audit retention, OR accountability

3.3 Source Requirements Traceability

WMSU TOR VI.E.3 compliance matrix:

TOR Requirement	Feature IDs	Implementation Notes	Compliance Status
Budget management and monitoring	F-BDG-01..08	Planning, obligations, availability checks, utilization	Delivered
Journal entry and general ledger	F-ACC-01..09	JEV, GL posting, trial balance, reconciliation, period close	Delivered
Tuition fee management	F-TUI-01..06	Fee schedules, payment policies, assessments	Delivered
Cashiering and receivables	F-CSH-01..10	Payments, OR issuance, daily collection, deposits	Delivered
Accounts payable and disbursement	F-APY-01..07	Invoices, DVs, checks/ADA, aging	Delivered

Procurement workflows	F-PRC-01..09	PR, RFQ, PO, IAR, vendor management	Delivered
Point-of-sale operations	F-POS-01..09	Terminals, sales, EOD, UIMS integration	Delivered
USiS integration	F-INT-01..06	Assessment handoff, payment status, clearance	Delivered
Financial reporting	F-RPT-01..08	Trial balance, budget vs actual, aging, SOA, export audit	Delivered
Configuration and setup	F-CFG-01..07	COA, funds, periods, OR series, workflows	Delivered

Deferred/Out of scope:

- Student registration and admissions → USiS responsibility
- TOR issuance → TORMS responsibility
- Inventory masters → UIMS responsibility
- Library operations → ULMIS

---

4. Non-Functional Requirements Summary

4.1 Performance Targets

Metric	Target	Measurement
API response (p95)	≤2s standard queries	APM monitoring
Payment + OR	≤3s	Performance profiling
Budget availability check	≤1s	Real-time testing
Trial balance (10K txns)	≤30s	Job duration metrics

4.2 Security Controls

Control	Implementation	Verification
TLS enforcement	Reverse proxy/ingress	SSL Labs scan
RBAC	`finance.*` permission checks	Automated security tests
Service keys	Internal API protection	Integration tests
Encryption at rest	Bank accounts, OR numbers	Security audit
Audit logging	All transactions, posting, config	Log review

OR accountability	Cashier assignment	Reconciliation reports

4.3 Reliability Measures

Measure	Target	Method
Availability	≥99.5% business hours	SLA monitoring
Backup frequency	Daily incremental, weekly full	Automated verification
RPO	≤15 minutes	DR drill testing
GL immutability	Reversing JEV only	Business rule enforcement
Period close integrity	Posting prevention on closed	Automated tests

---

5. Actors & Feature Access

Actor	Primary Features	Restricted Features
Cashier	F-CSH-01..10, F-POS-01..09	Own cashier operations
Accountant	F-ACC-01..09, F-APY-01..07, F-RPT-01, F-RPT-04	Full accounting
Budget Officer	F-BDG-01..08, F-RPT-02	Budget oversight
Procurement Officer	F-PRC-01..09	Procurement processes
Treasury	F-APY-03..06, F-ACC-06	Disbursement authority
Finance Admin	F-CFG-01..07, All reports	System configuration
USiS (System)	F-INT-01..06	Internal API only (service key)
TORMS (System)	F-INT-05..06	Clearance API only (service key)

End of Features & Requirements — UFMS

Part 4 — Low-Level Design (LLD) Workflows

UFMS — University Financial Management System Low-Level Design

Field	Value
Document ID	LLD-UFMS-001
Product	UFMS
Companion	UFMS-SRS.md
Date	November 2026

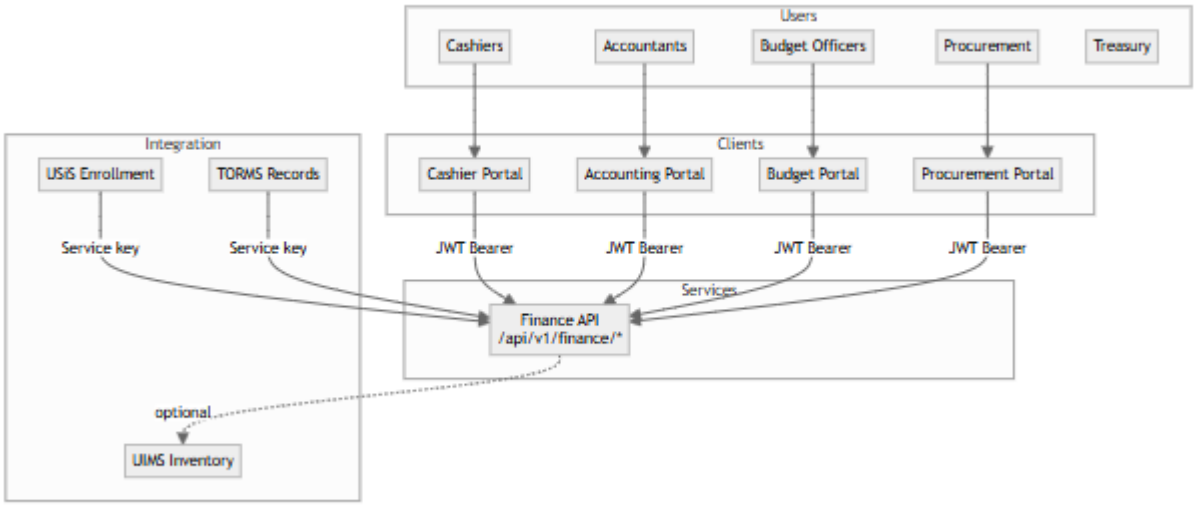
---

1. Purpose

This Low-Level Design documents critical workflows for UFMS cashiering, budget management, procurement, disbursement, POS operations, and USiS integration — providing sequence diagrams, state machines, and implementation details.

---

2. System Context (Multi-Role Architecture)



Role-specific routes:

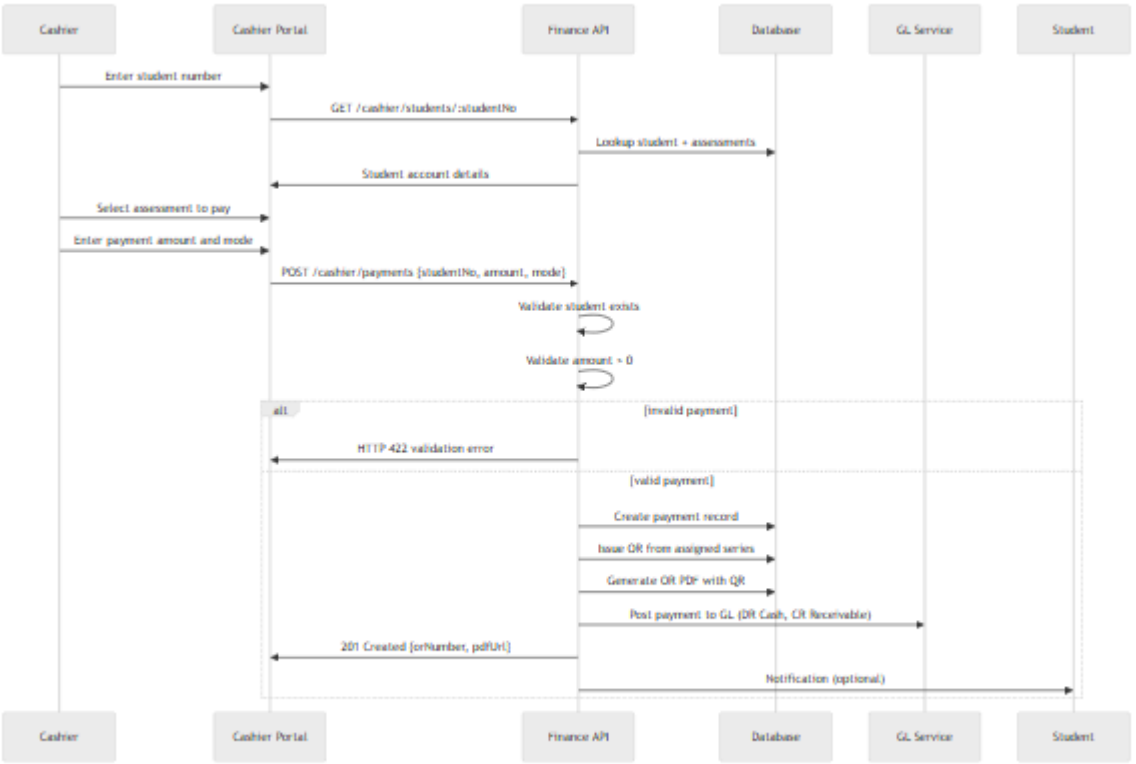
- Cashier: `/cashier/*`
- Accounting: `/accounting/*`

- Budget: `/budget/*`
- Procurement: `/procurement/*`
- Treasury: `/treasury/*`

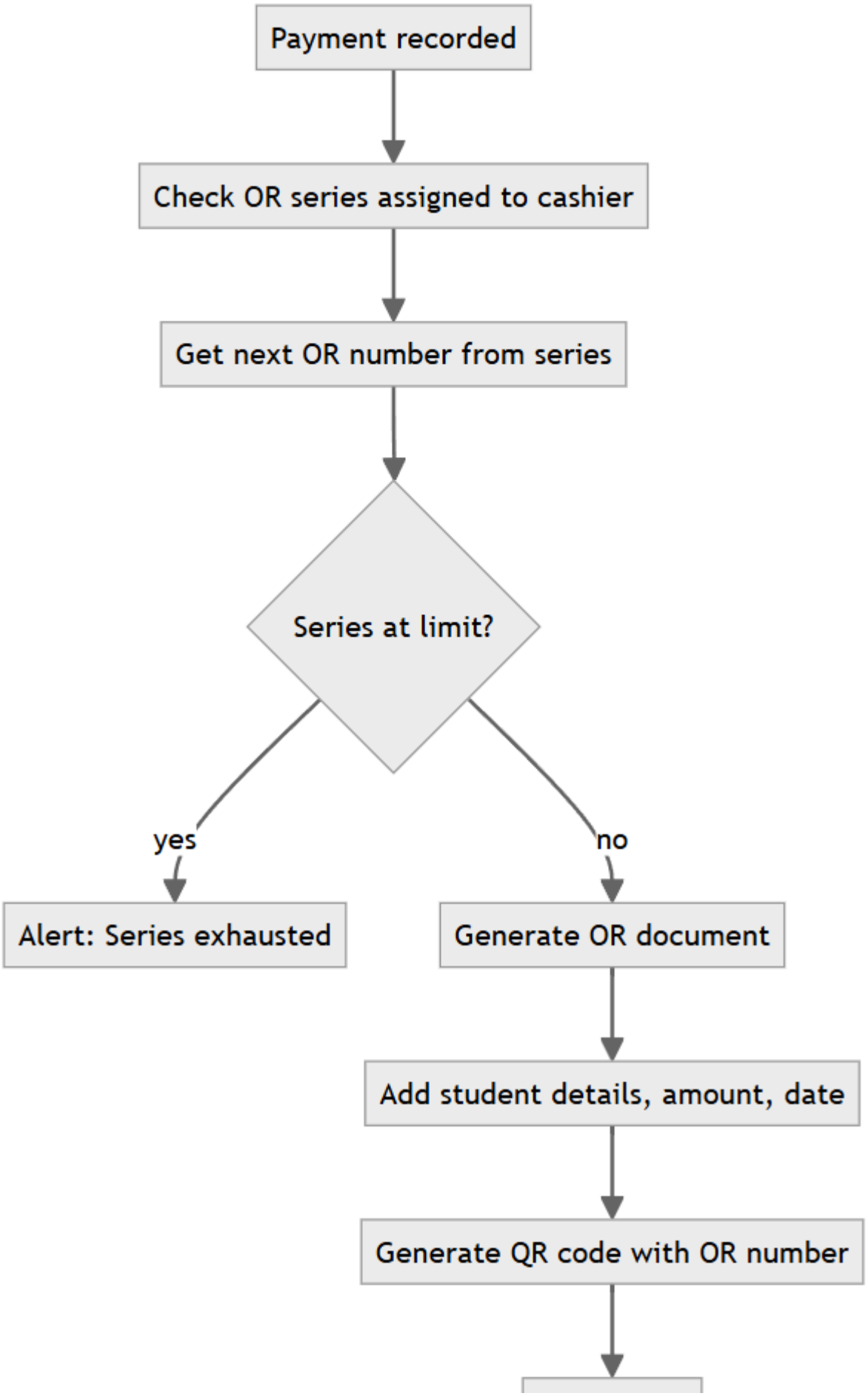
---

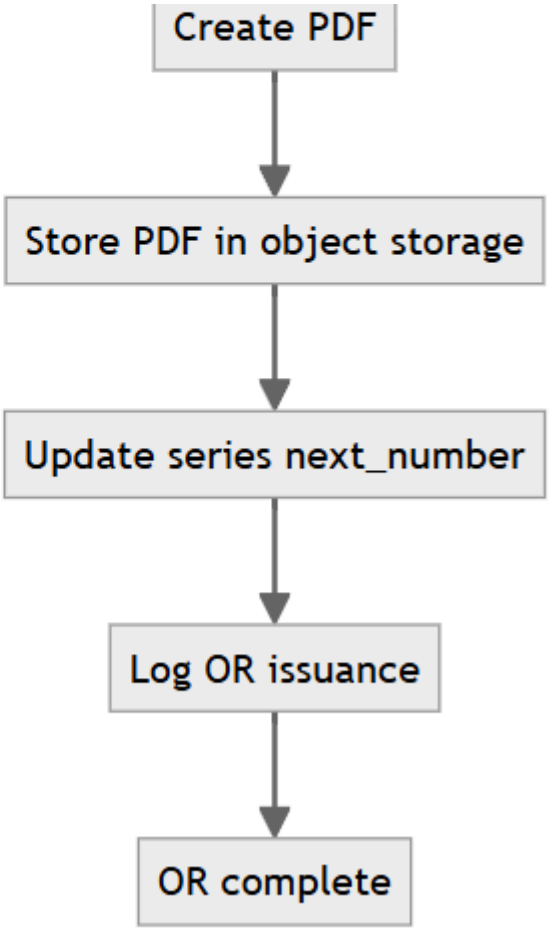
3. Cashiering Workflow

3.1 Student Payment Processing

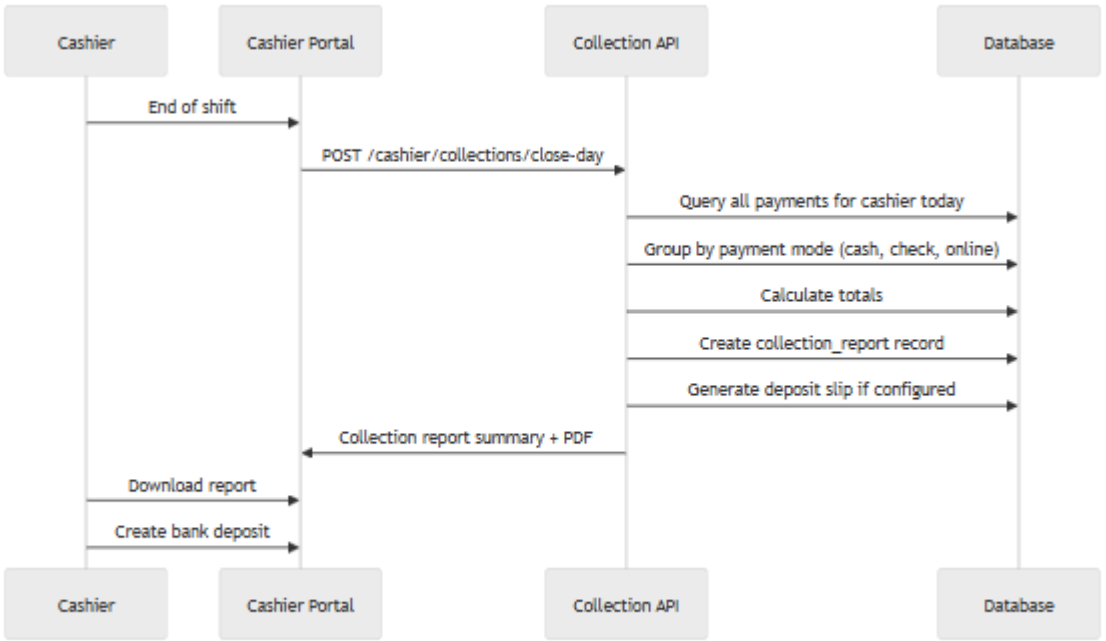


3.2 Official Receipt Issuance


Payment recorded
Check OR series assigned to cashier
Get next OR number from series
Series at limit?
yes
no
Alert: Series exhausted
Generate OR document
Add student details, amount, date
Generate QR code with OR number

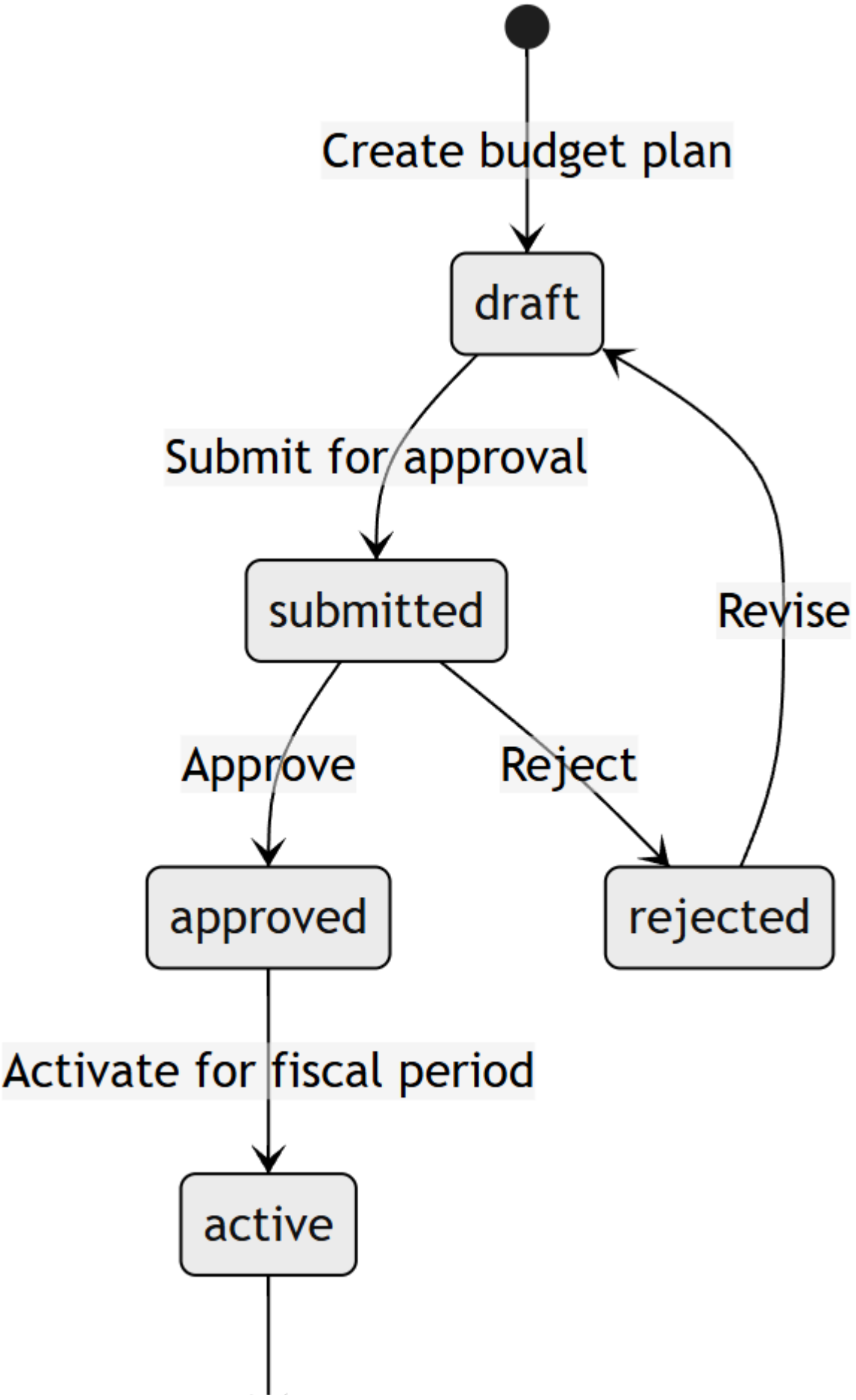


3.3 Daily Collection Report

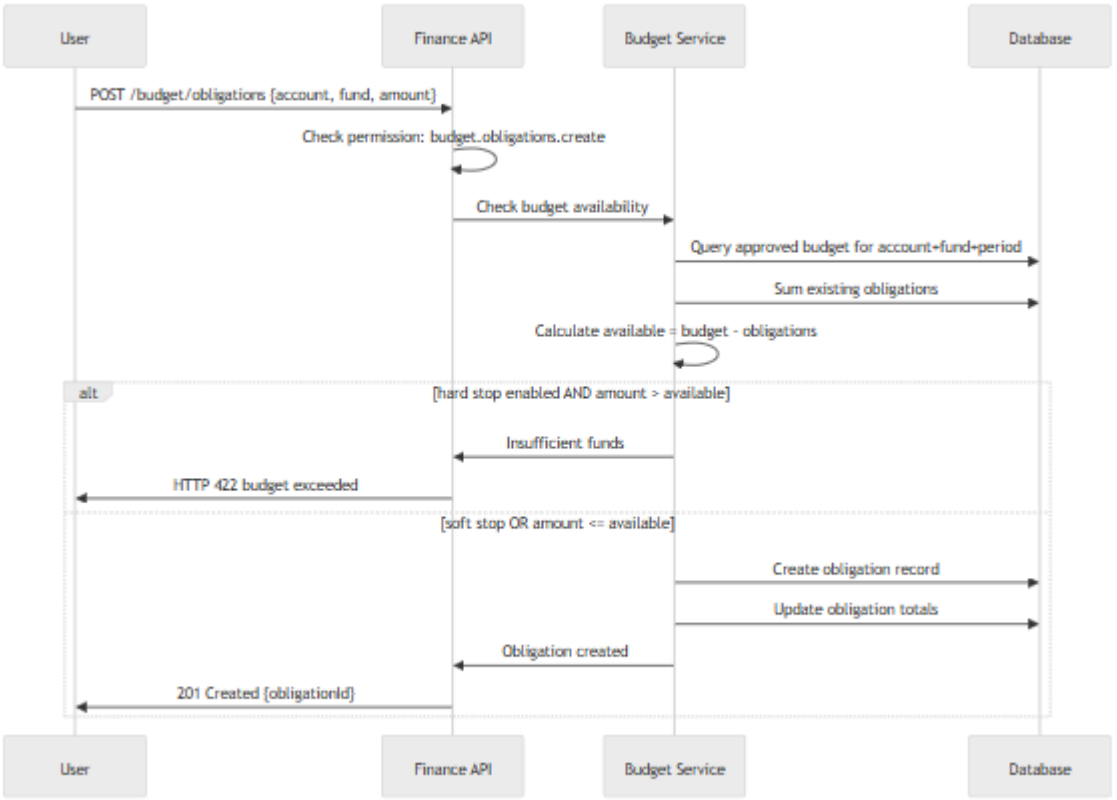


4. Budget Workflow

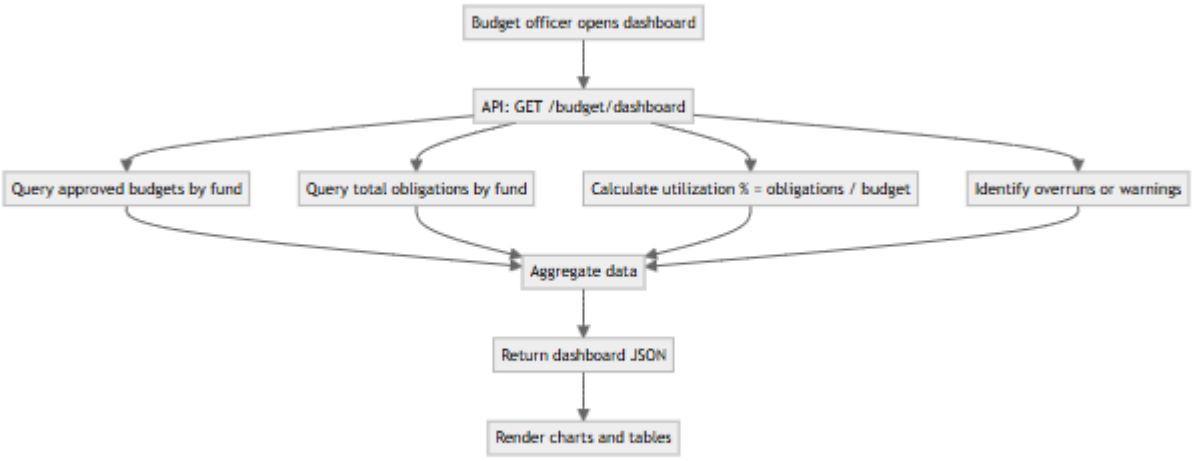
4.1 Budget Planning and Approval


Create budget plan
draft
Submit for approval
submitted
Revise
Approve
Reject
approved
rejected
Activate for fiscal period
active

4.2 Budget Obligation and Availability Check



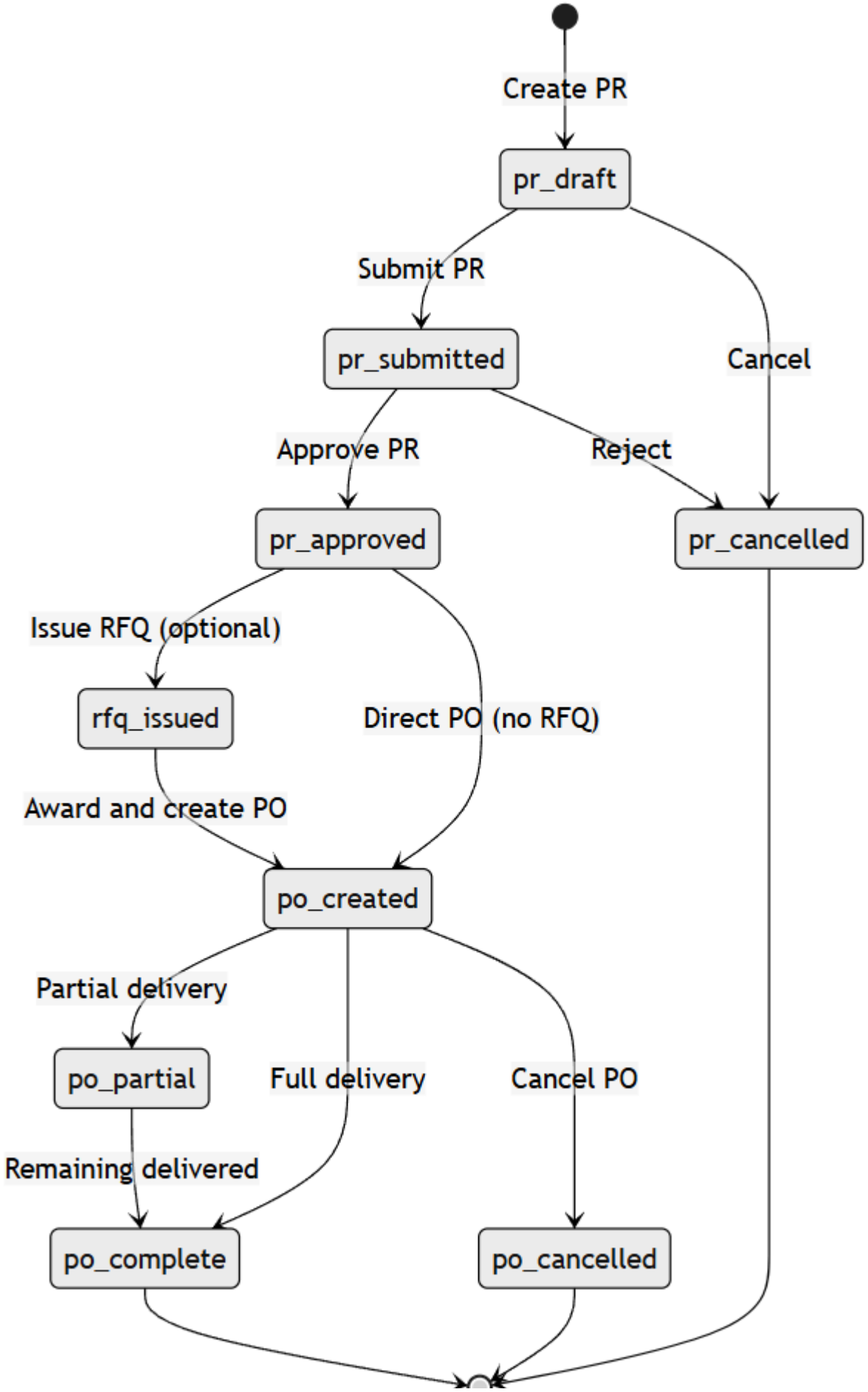
4.3 Budget Monitoring Dashboard



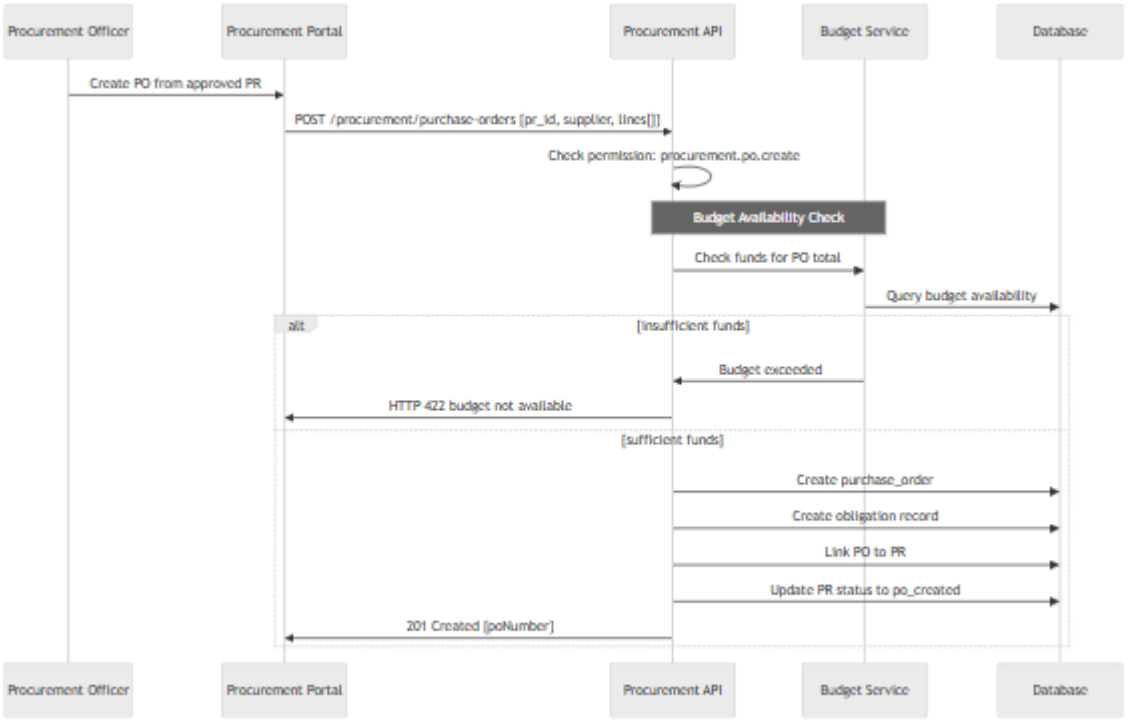
---

5. Procurement Workflow

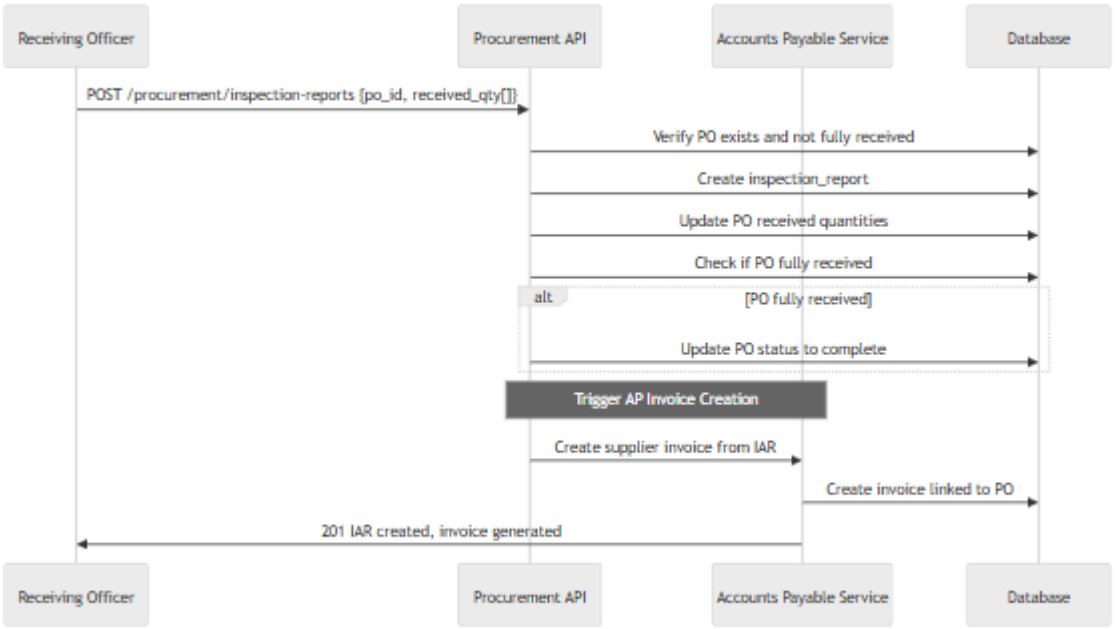
5.1 Procurement Lifecycle State Machine


Create PR
pr_draft
Submit PR
pr_submitted
Cancel
Approve PR
Reject
pr_approved
pr_cancelled
Issue RFQ (optional)
rfq_issued
Direct PO (no RFQ)
Award and create PO
po_created
Partial delivery
po_partial
Full delivery
Cancel PO
Remaining delivered
po_complete
po_cancelled

5.2 Purchase Order Creation and Budget Check



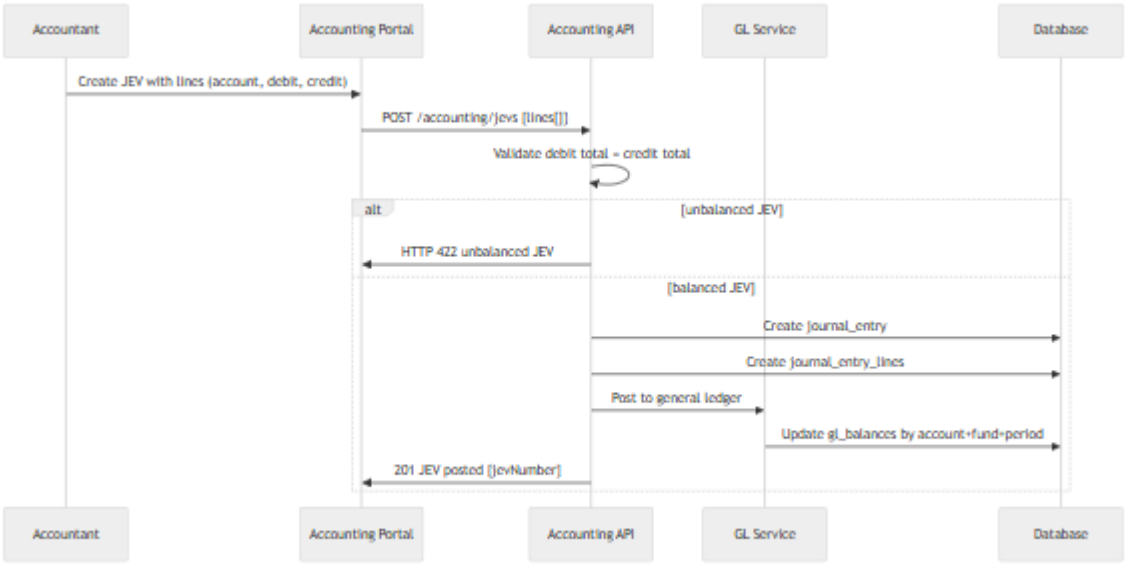
5.3 Inspection and Acceptance Report (IAR)



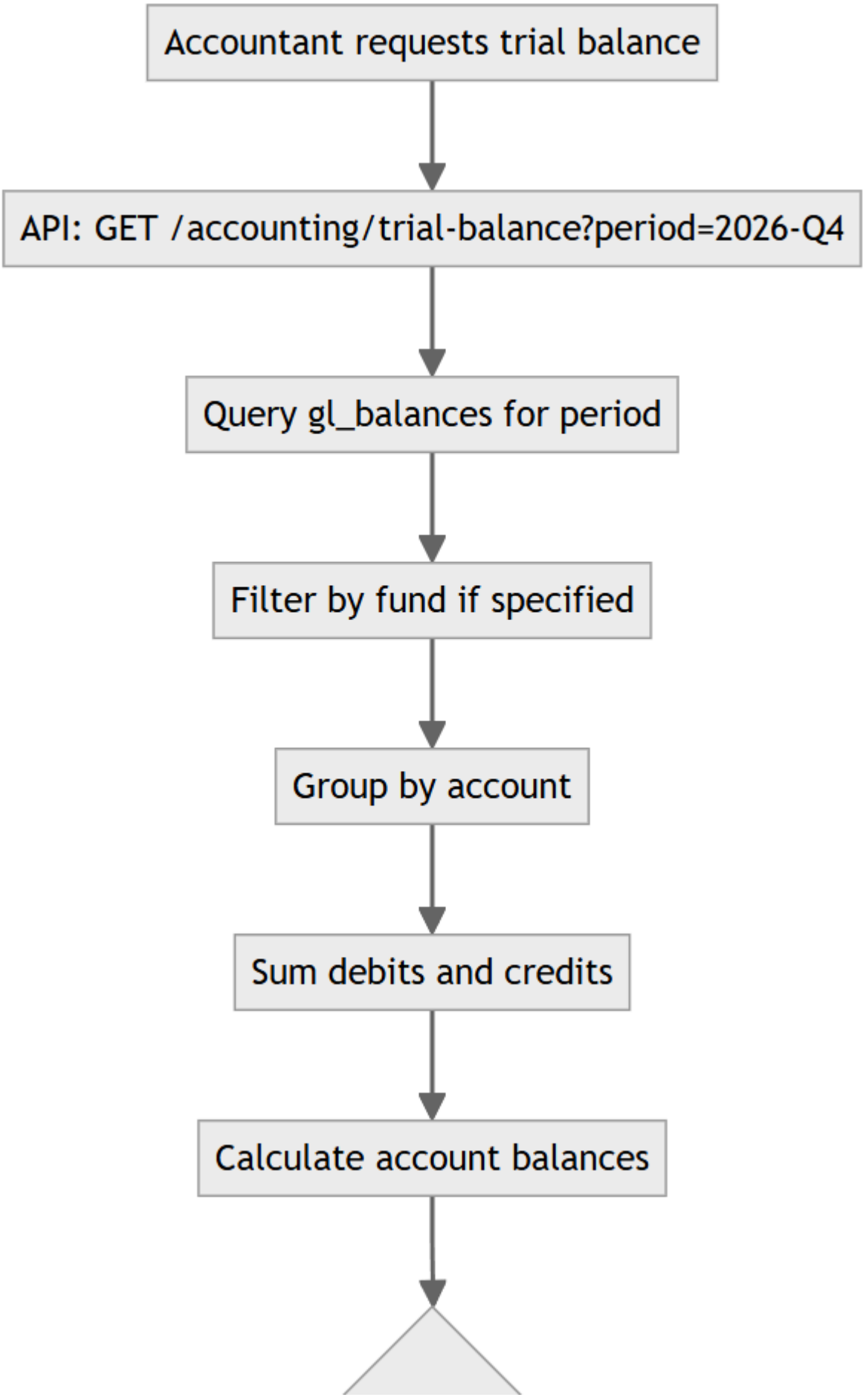
---

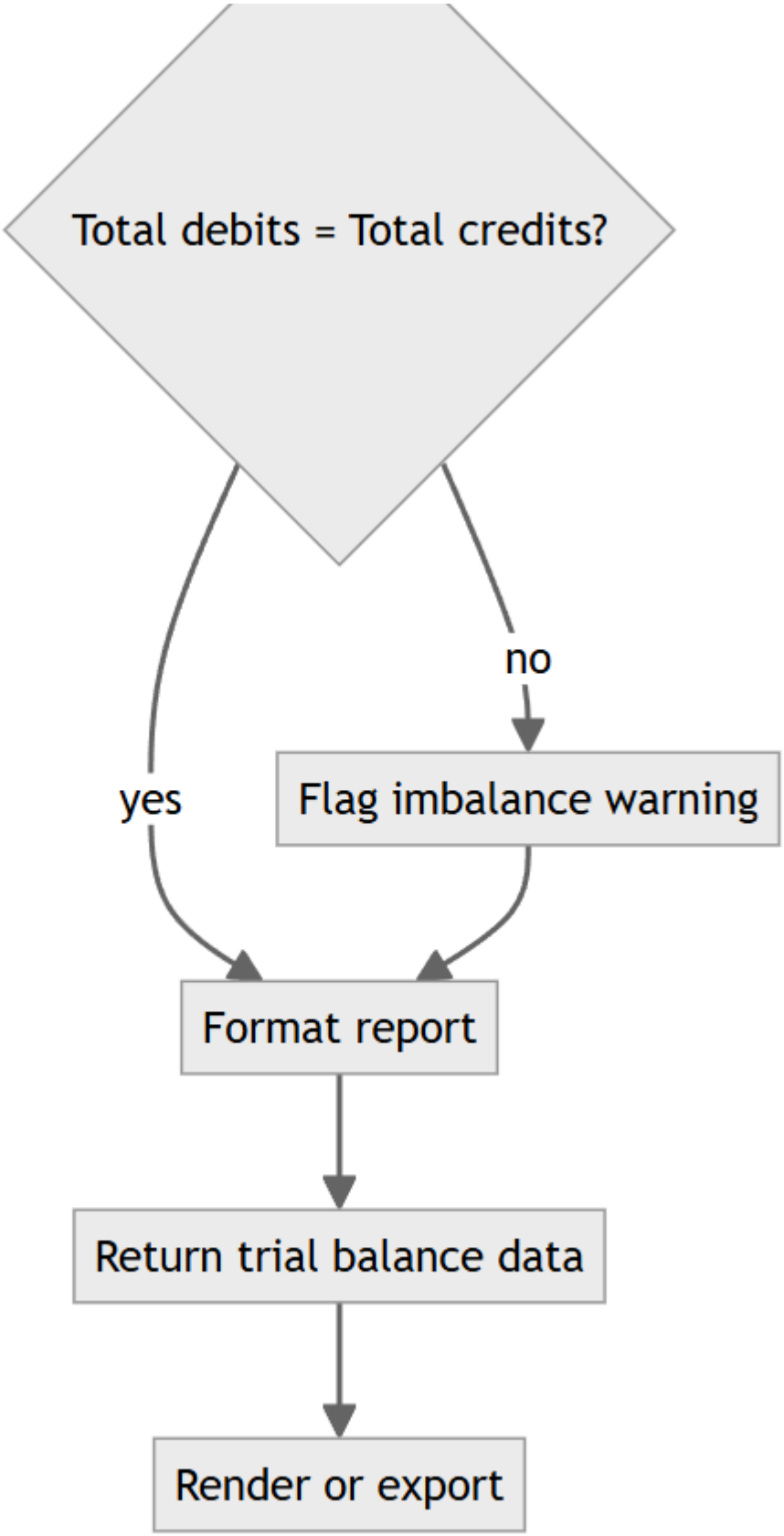
6. Accounting Workflow

6.1 Journal Entry Voucher (JEV) Posting

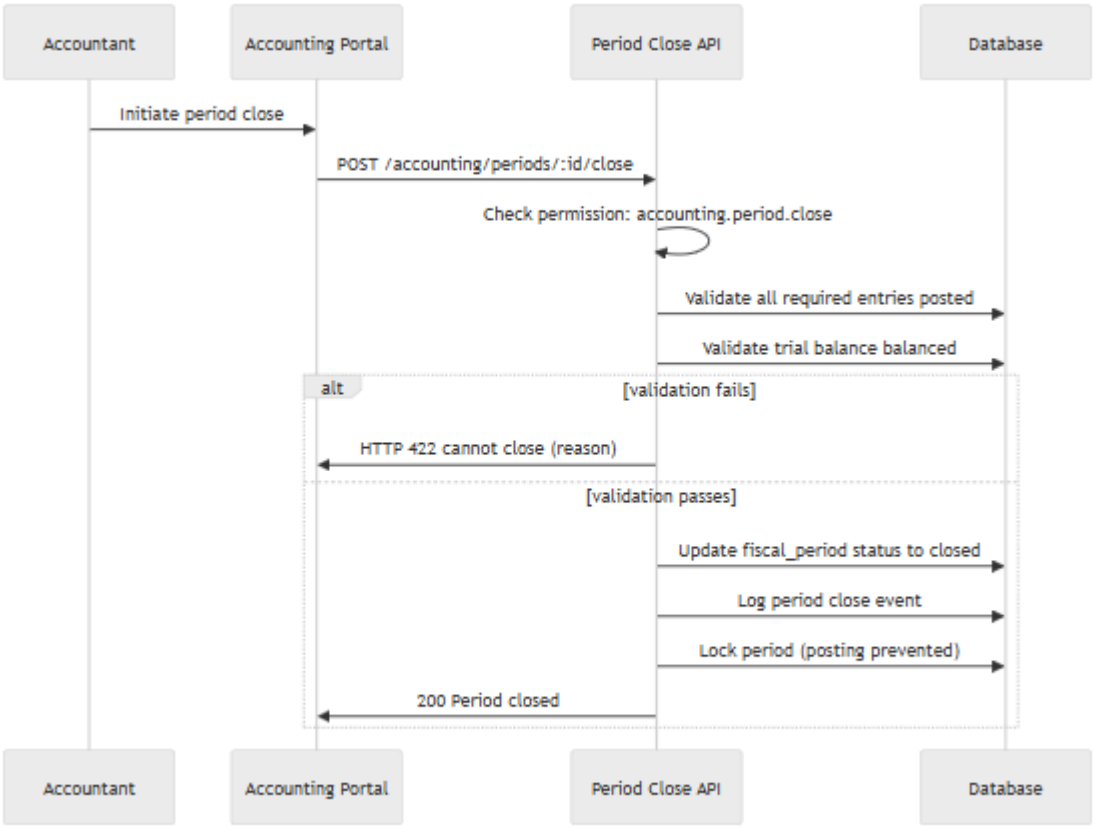


6.2 Trial Balance Generation


Accountant requests trial balance
API: GET /accounting/trial-balance?period=2026-Q4
Query gl_balances for period
Filter by fund if specified
Group by account
Sum debits and credits
Calculate account balances



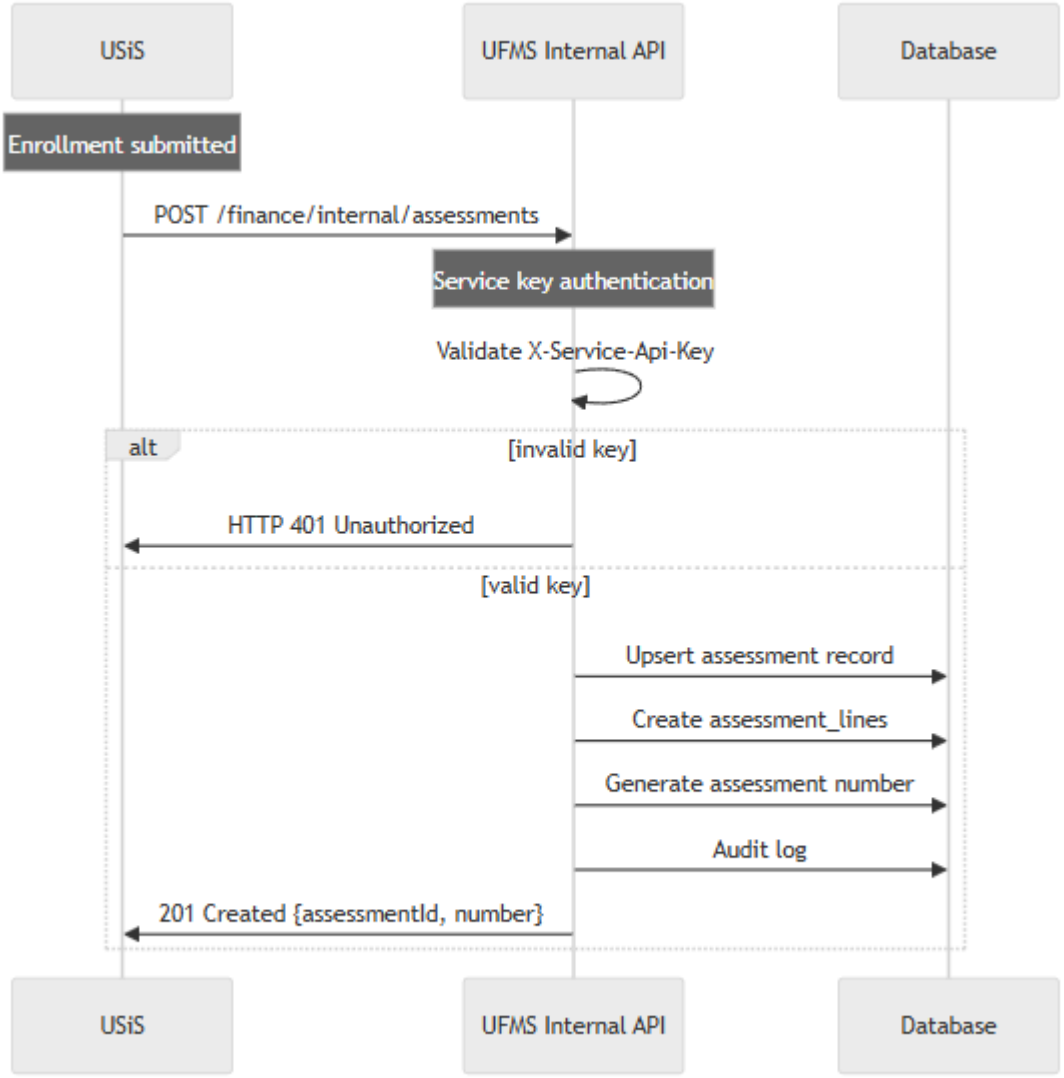
6.3 Fiscal Period Close



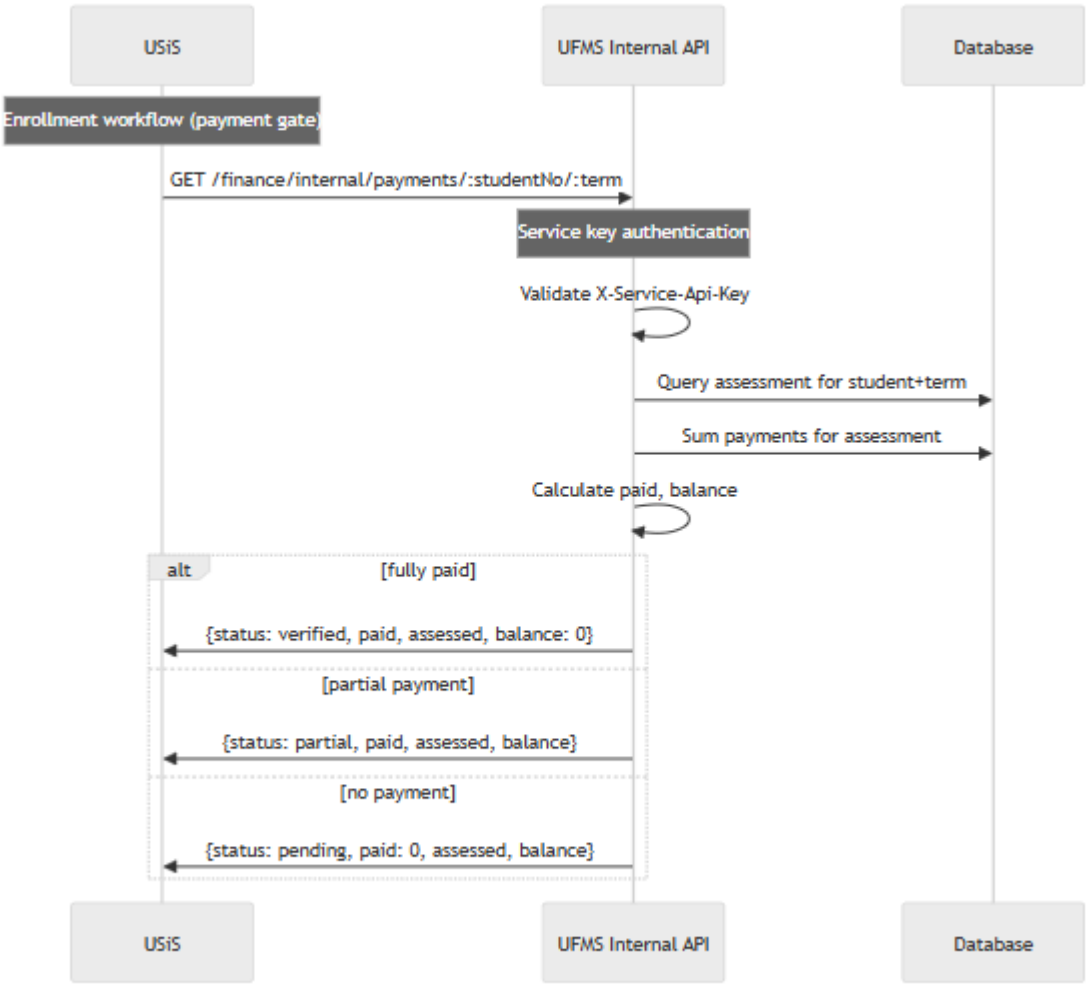
---

7. USiS Integration Workflows

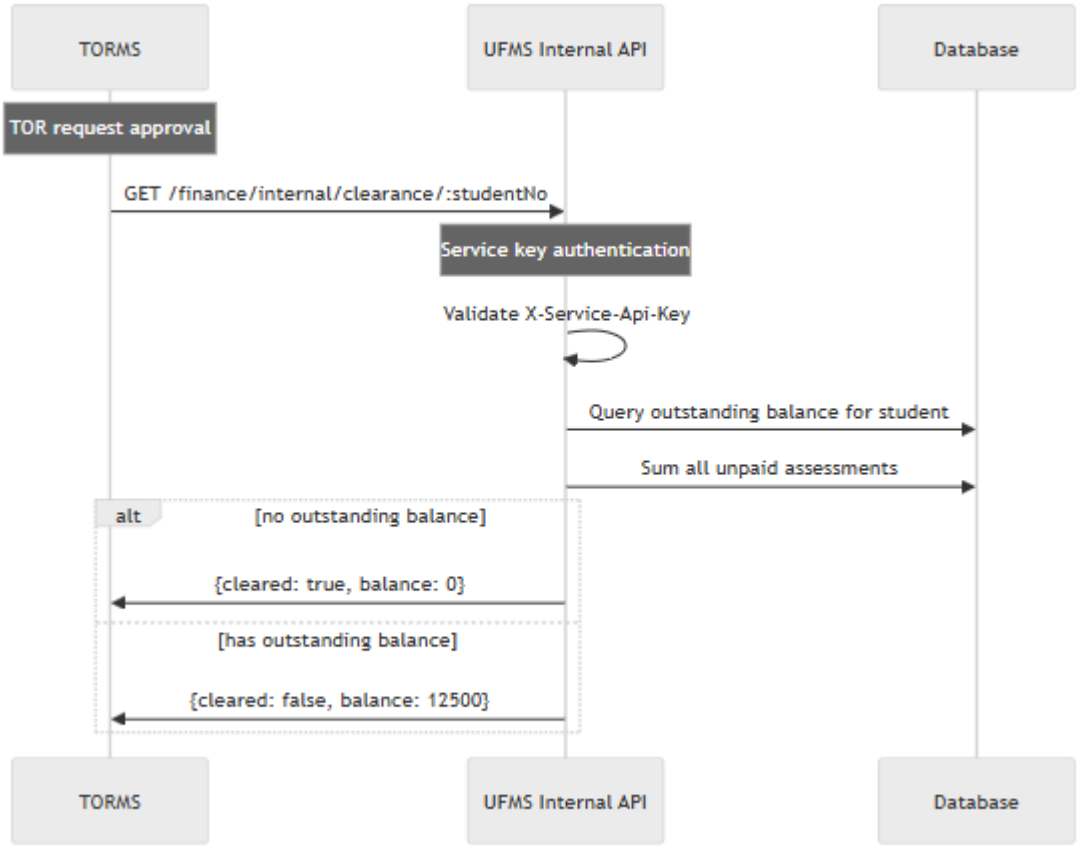
7.1 Assessment Upsert from USiS



7.2 Payment Status Query from USiS



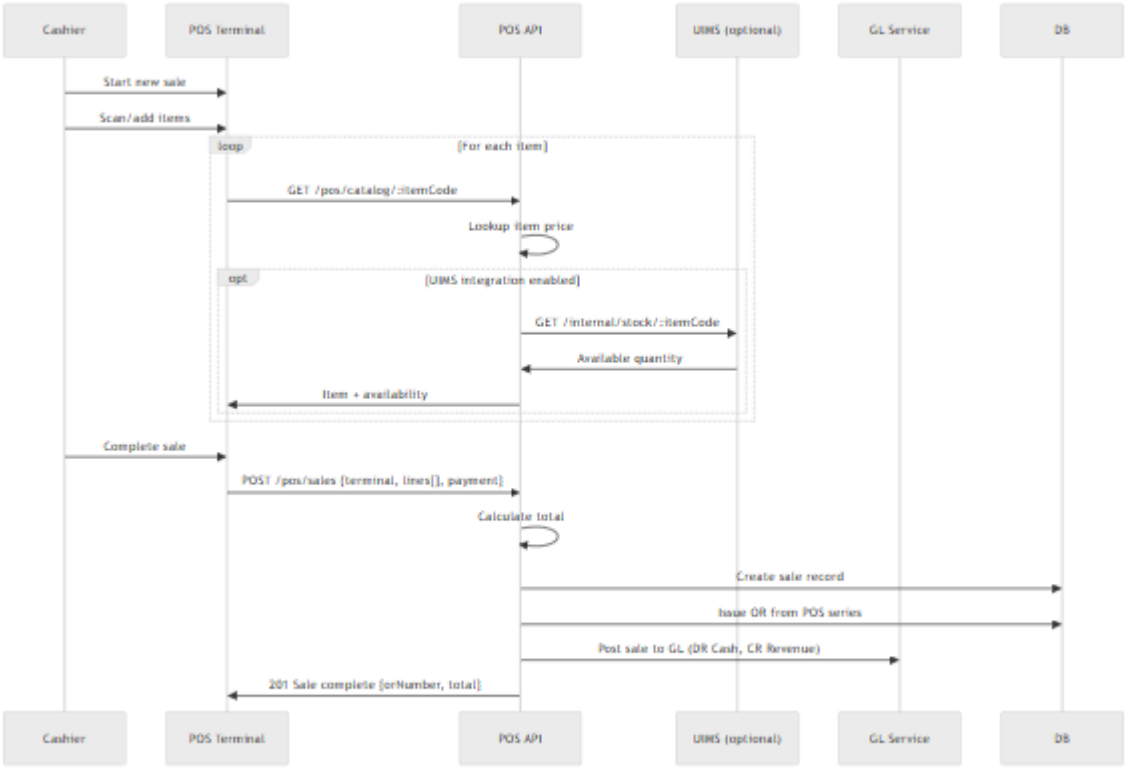
7.3 Clearance Check from TORMS



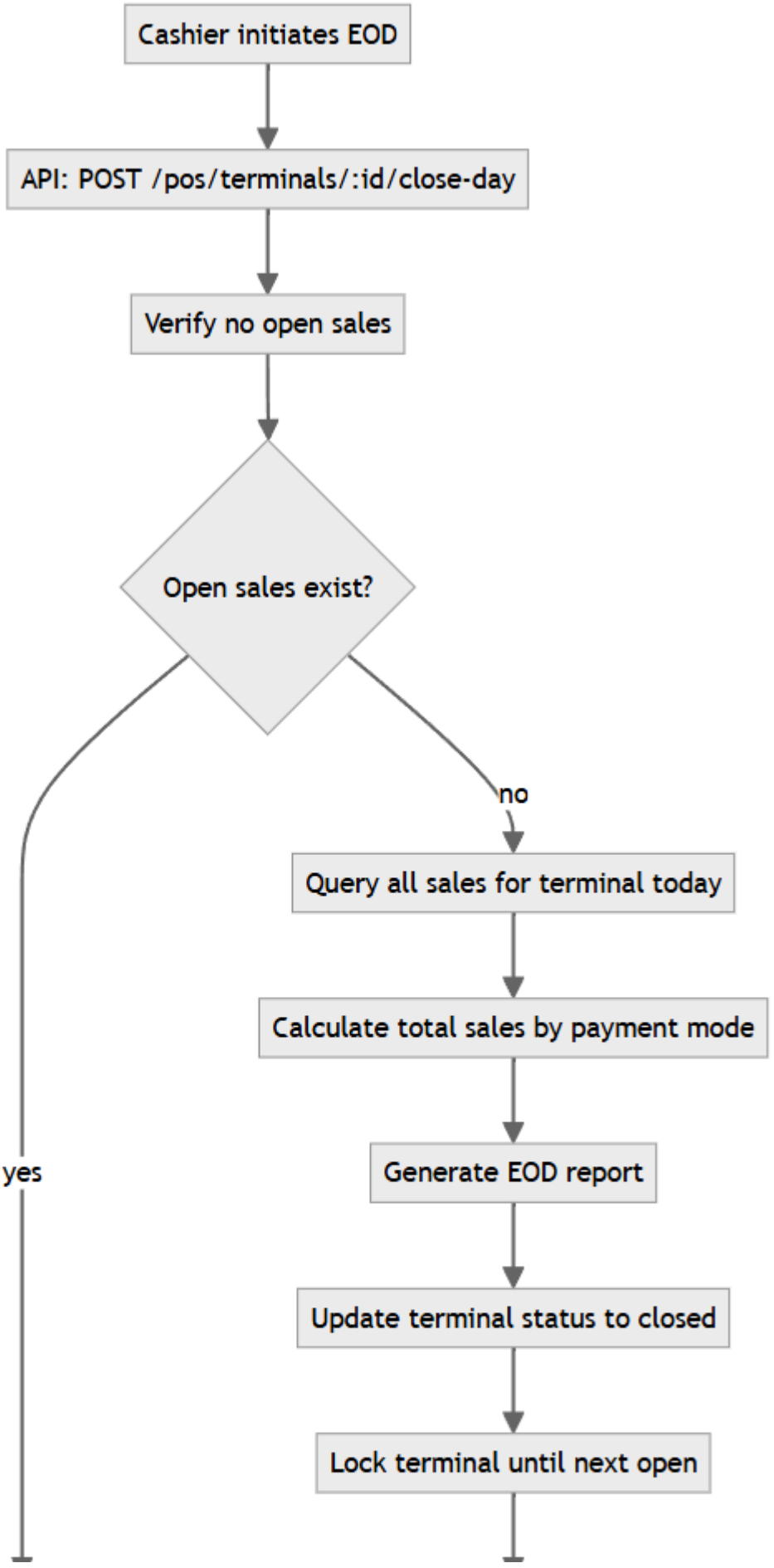
---

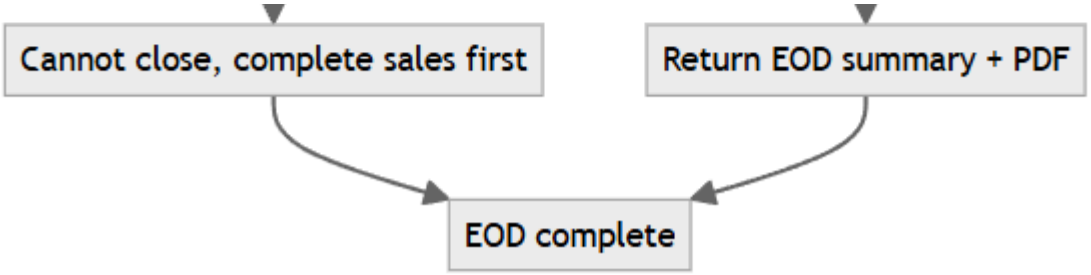
8. Point-of-Sale (POS) Workflow

8.1 POS Sale Transaction



8.2 POS End-of-Day Closing

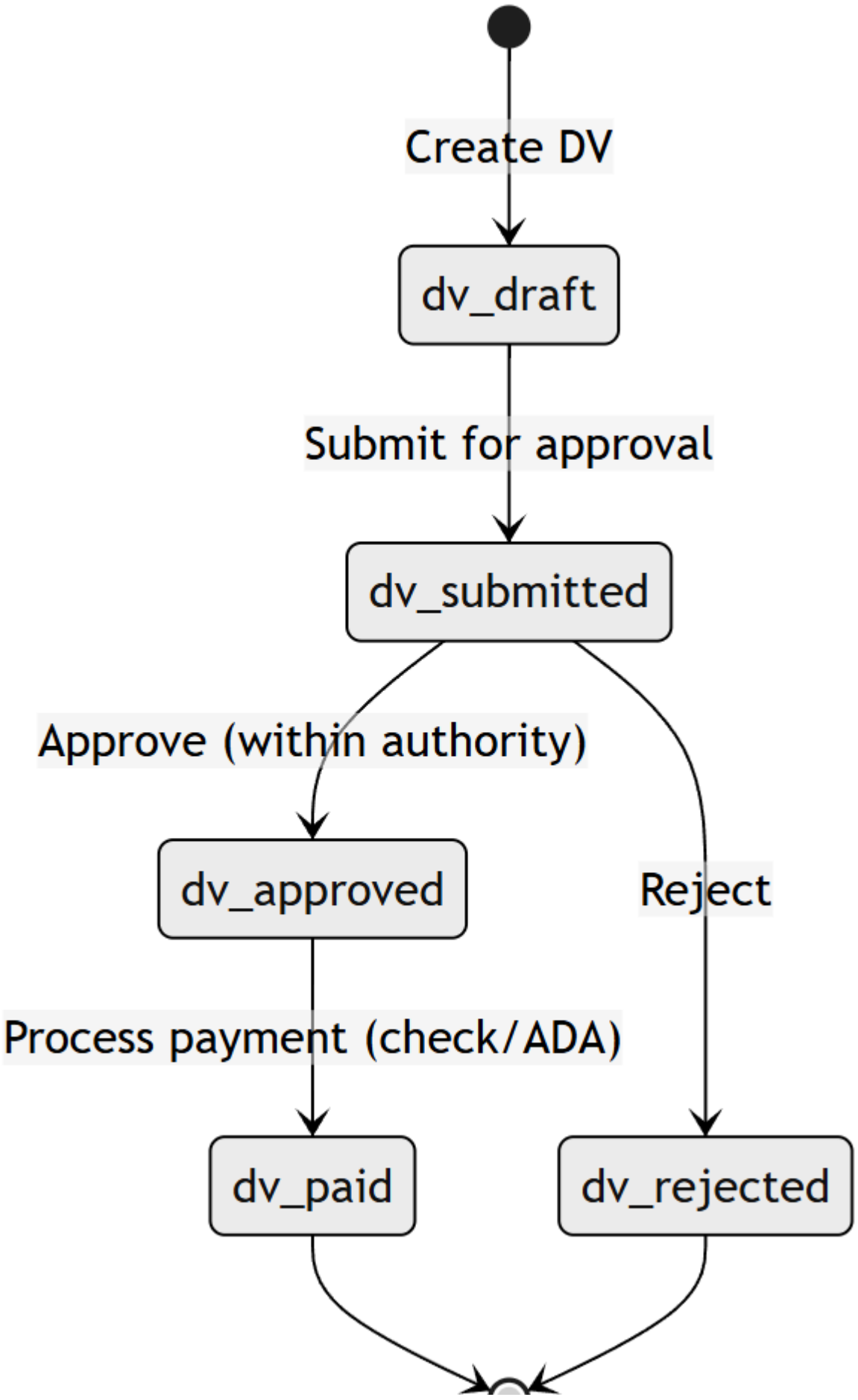

Cashier initiates EOD
API: POST /pos/terminals/:id/close-day
Verify no open sales
Open sales exist?
yes
no
Query all sales for terminal today
Calculate total sales by payment mode
Generate EOD report
Update terminal status to closed
Lock terminal until next open



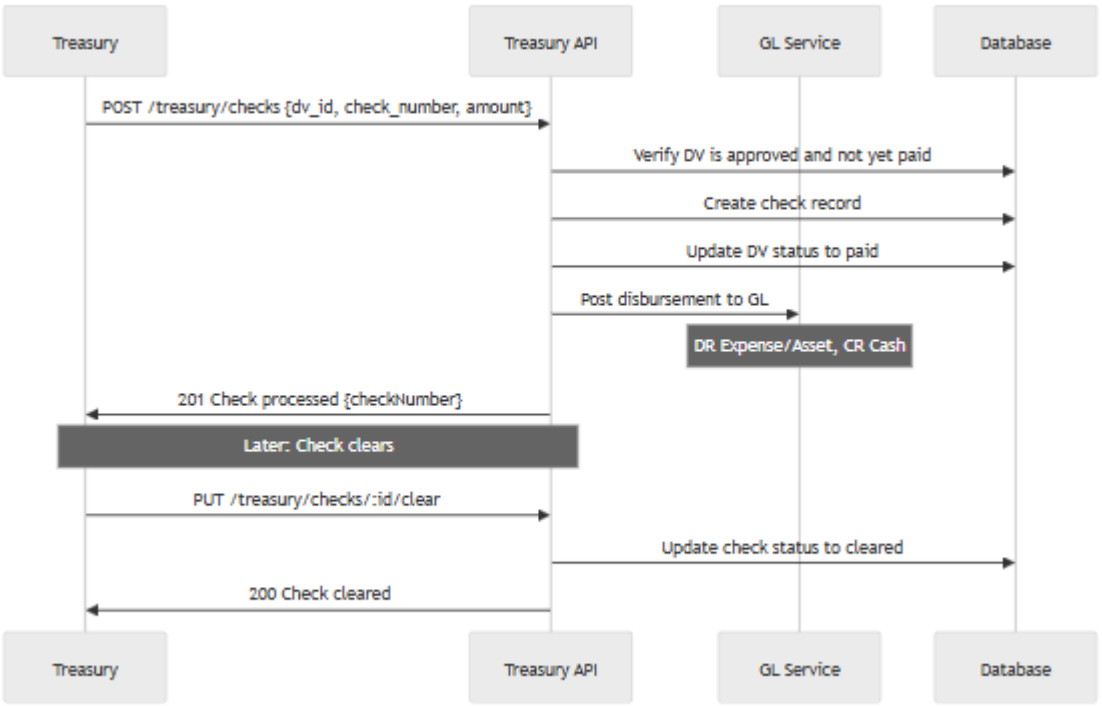
---

9. Disbursement Workflow

9.1 Disbursement Voucher (DV) Processing


Create DV
dv_draft
Submit for approval
dv_submitted
Approve (within authority)
dv_approved
Reject
Process payment (check/ADA)
dv_paid
dv_rejected

9.2 Check Processing



---

10. Error Cases & Edge Conditions

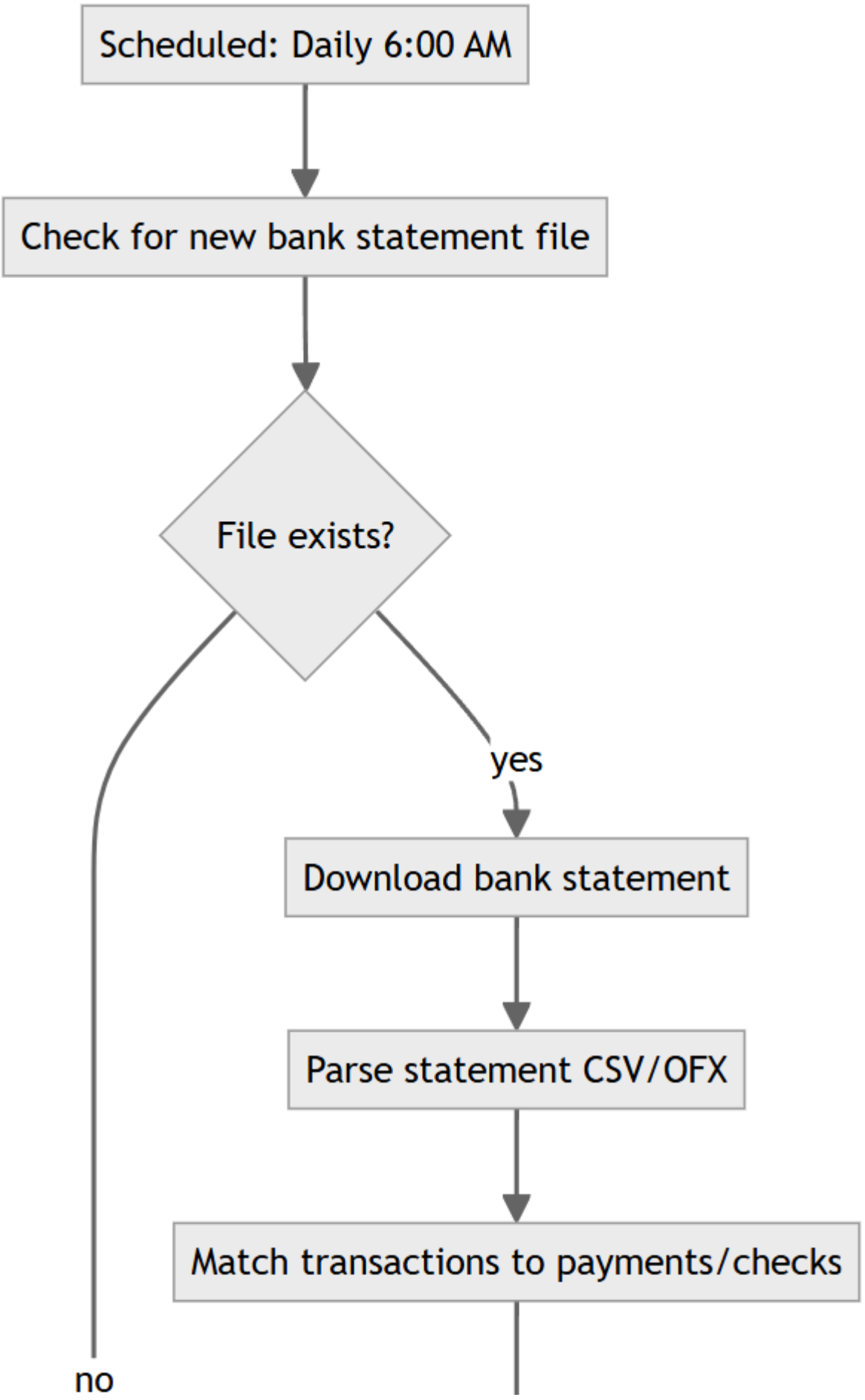
Case	Expected Behavior	HTTP Status	User Message
Record payment for unknown student	Reject payment	422	"Student not found."
Record payment with amount ≤ 0	Reject payment	422	"Invalid payment amount."
Budget obligation exceeds available (hard stop)	Block obligation	422	"Insufficient budget: Available ₱50,000, Requested ₱75,000."
Post JEV with unbalanced debit/credit	Block posting	422	"JEV unbalanced: Debit ₱100,000, Credit ₱95,000."
Post to closed fiscal period	Block posting	422	"Cannot post to closed period 2026-Q3."

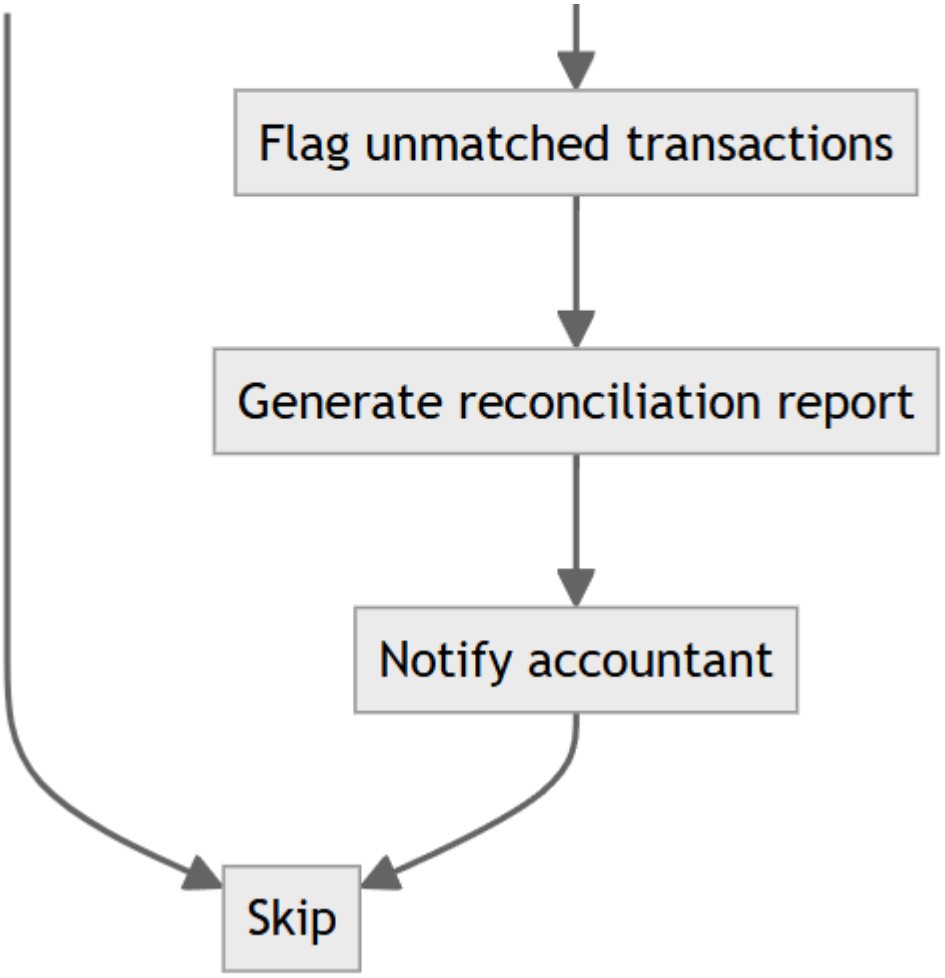
USiS calls internal API without service key	Block request	401	"Service key required."
Create PO without approved PR	Block PO creation	422	"PR must be approved before PO."
IAR for non-existent PO	Reject IAR	404	"Purchase order not found."
POS sale on closed terminal	Block sale	422	"Terminal is closed. Start new day first."
Close EOD with open sales	Block close	422	"Cannot close. {N} sales still open."
OR series exhausted	Alert + block payment	422	"OR series exhausted. Request new series."
UIMS unavailable during POS stock check	Log warning, allow sale without stock check	200	"(Stock check unavailable)"

---

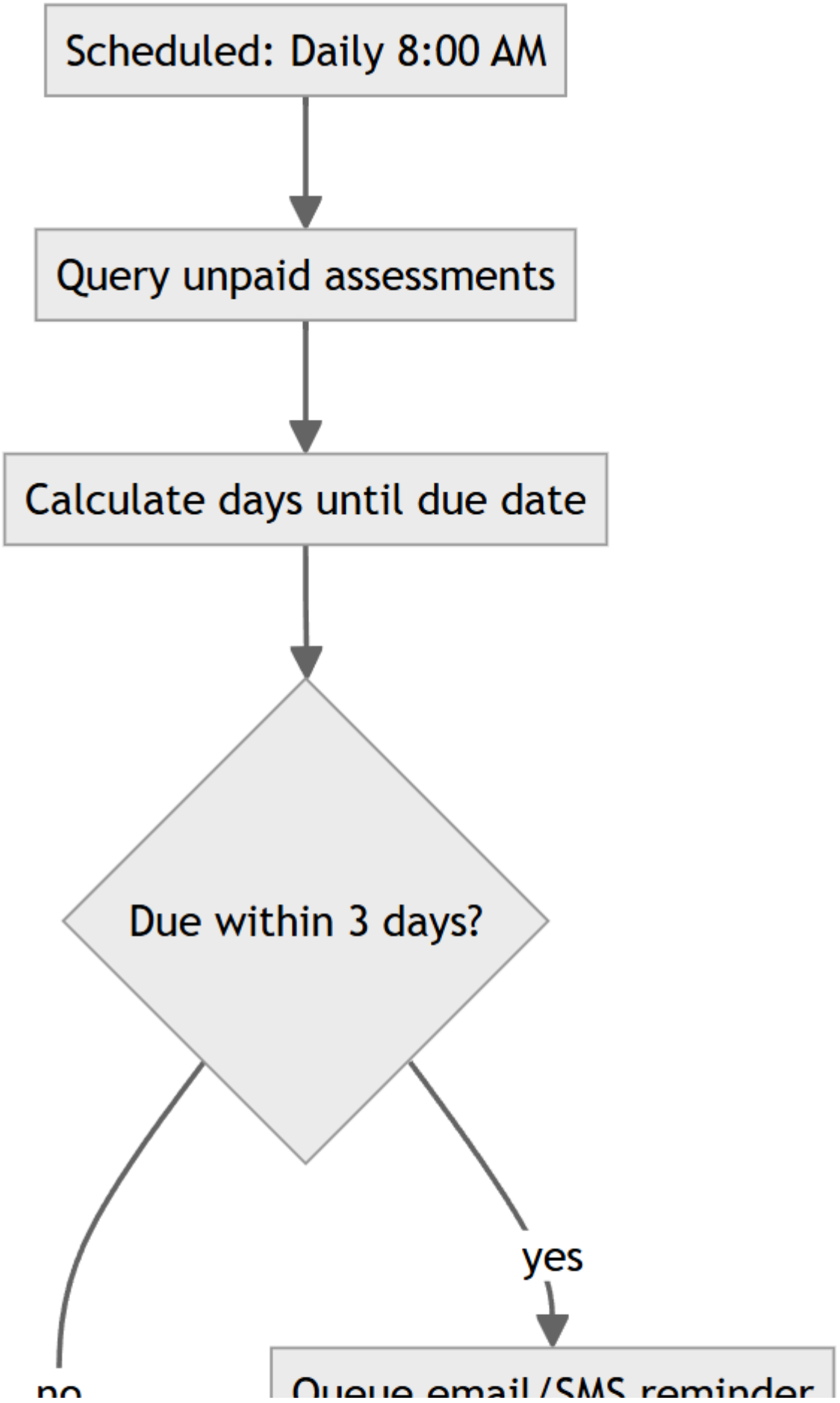
11. Background Job Workflows

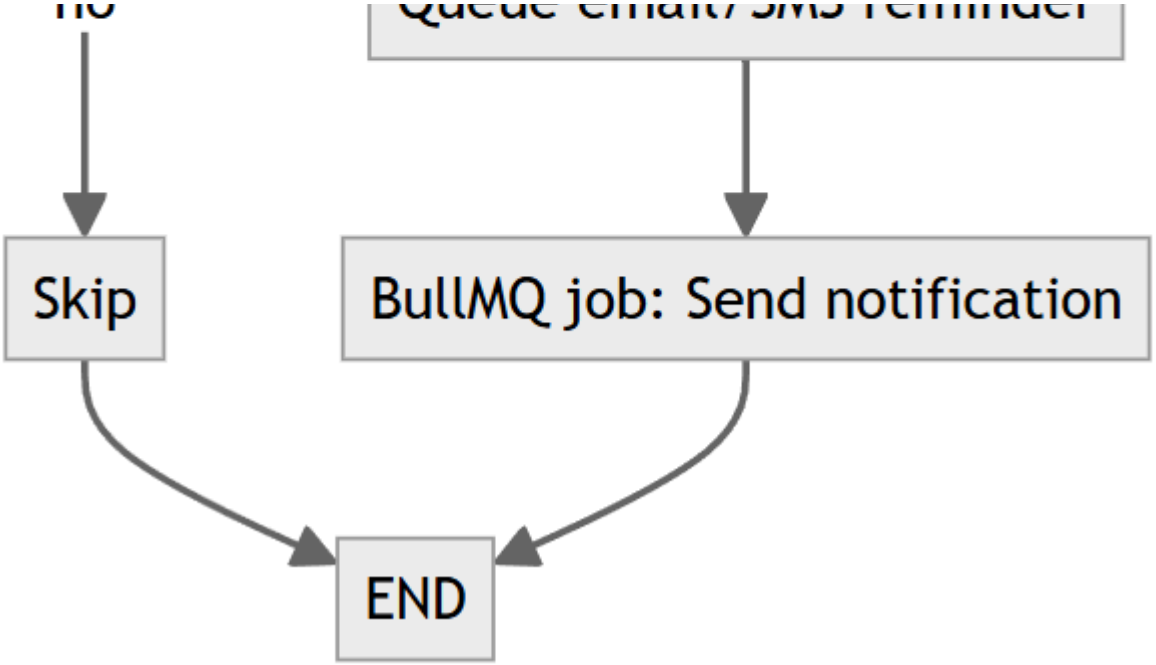
11.1 Bank Reconciliation Import


Scheduled: Daily 6:00 AM
Check for new bank statement file
File exists?
yes
Download bank statement
Parse statement CSV/OFX
Match transactions to payments/checks
no



11.2 Payment Reminder Notifications


Scheduled: Daily 8:00 AM
Query unpaid assessments
Calculate days until due date
Due within 3 days?
yes
no
Queue email/SMS reminder



End of LLD — UFMS
